

THE UNIVERSITY OF THE WEST INDIES



B.Sc. (Engineering)
Department of Electrical and Computer
Engineering
ECNG 3020 – SPECIAL PROJECT

**Optimal Power Sharing in DC Grid-
Connected Microgrids for Rural
Electrification**

System Testing Document
(STD)

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Project Type: IV

ELECTRICAL
& COMPUTER
ENGINEERING
DEPARTMENT

Prepared by:

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Based on the ISO/IEC/IEEE Std 29148:2018

Statement of Academic Honesty

Appendix C: Statement of Academic Honesty

(Word™ version available on myelearning)



THE UNIVERSITY OF THE WEST INDIES
ST. AUGUSTINE, TRINIDAD & TOBAGO, WEST INDIES
FACULTY OF ENGINEERING
Department of Electrical & Computer Engineering
B. Sc. in Electrical & Computer Engineering

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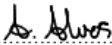
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Document Control

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Version: 1.0

Date: 26/03/2026

Author(s): Aaron Alves

Document Signoff

Nature of Signoff	Person	Signature with Date	Role
Author	Aaron Alves		Project Manager, Requirements Analyst and Consultant
Reviewees			QA Officer

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Document Change Record

Date	Version	Author	Change Details
27 Sept 2022	0.0	Azim Abdool	Initial template commit.
3 rd February 2026	1.0	Aaron Alves	Defined Introduction, System Overview and Test Environment
26 th March 2026	1.0	Aaron Alves	Completed Testing Strategies

1. Introduction

This document presents the system testing carried out for the DC microgrid optimal power flow (OPF) framework. The testing evaluates the system's performance across multiple test cases, ensuring that the optimization model operates correctly, satisfies constraints, and produces reliable results under different operating conditions.

2. System Overview

The system is a MATLAB-based backend framework designed to model and optimize power flow in DC microgrids for rural electrification. It performs optimal power flow (OPF) analysis to manage distributed energy resources, including photovoltaic generation, battery storage, and grid interaction. The system supports multiple test cases and operates over a 24-hour time horizon, enabling evaluation of power distribution, energy sharing, and system losses under both islanded and grid-connected conditions.

3. Test Environment

The testing of the system was carried out using a MATLAB-based computational environment. The optimization framework was developed and executed within MATLAB, utilizing built-in numerical solvers for nonlinear optimization.

The tools used for testing consist of:

- MATLAB (with Optimization Toolbox for fmincon)

The system was tested on a standard personal computer capable of handling iterative optimization routines and time-series simulations.

4. Test Systems

4.1 Radial 7-bus System

The system is modelled as a 7-bus DC microgrid representing a local community. Bus 1 serves as a central load, while the remaining buses are prosumer households with PV generation and battery storage. The network has a radial topology with voltage limits of 0.95–1.05 p.u. Each household includes a 4 p.u.-h battery with SOC limits of 0.50–0.90. Distribution lines are modelled as resistive branches with values between 0.04 and 0.06 p.u., typical of low-voltage DC networks.



Figure 1: Test Case 1: Radial 7-Bus System

4.2 IEEE 14-bus System

The system is modelled as a modified IEEE 14-bus network adapted for a DC microgrid. Since no standard DC benchmark exists, the AC system from MATPOWER is modified by removing reactive power and setting line reactance to zero. Thirteen buses represent prosumer

households with PV and battery storage, while Bus 14 serves as a central load. Line resistances are reduced by 10%, and transformer branches are replaced with small resistances (0.005 p.u.) to represent DC/DC converters. Voltage limits are set between 0.95–1.05 p.u., and each household includes a 4 p.u.-h battery with SOC limits of 0.50–0.90. This model enables analysis of power sharing, voltage regulation, and losses in a larger DC microgrid.

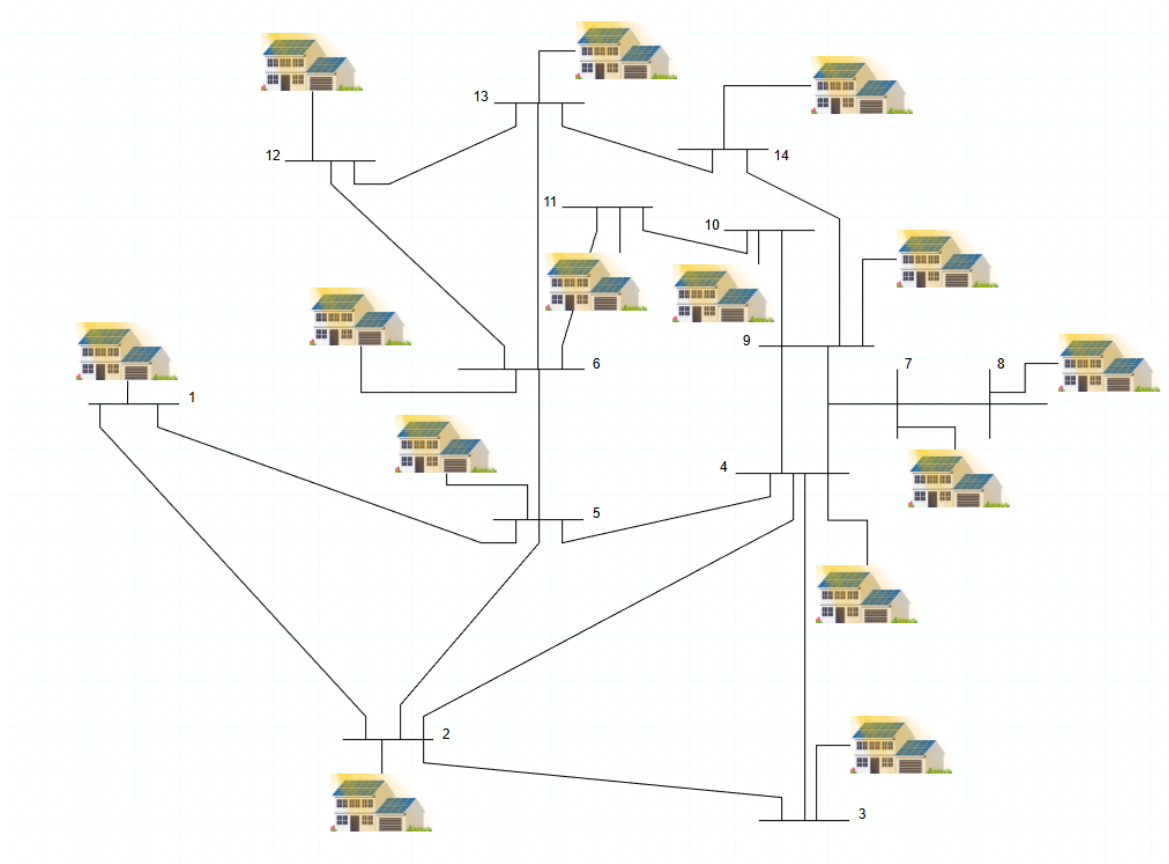


Figure 2: Test Case 2: IEEE 14-Bus System

4.3 Clustered 20-bus System

The system is modelled as a 20-bus DC microgrid representing interconnected households. Bus 20 serves as a central load, while the remaining buses are prosumer households with PV generation and battery storage. The network is divided into four clusters of five buses, each

arranged in a ring topology to enable local power sharing and improve resilience. Clusters are interconnected via gateway nodes, forming a higher-level ring for inter-cluster energy exchange. Each household includes a 4 p.u.-h battery with SOC limits of 0.50–0.90. Distribution lines are modelled as resistive branches, with 0.04 p.u. for intra-cluster and 0.06 p.u. for inter-cluster connections. Voltage limits are maintained between 0.95–1.05 p.u., enabling analysis of power sharing, losses, and system performance in a clustered DGDS microgrid.

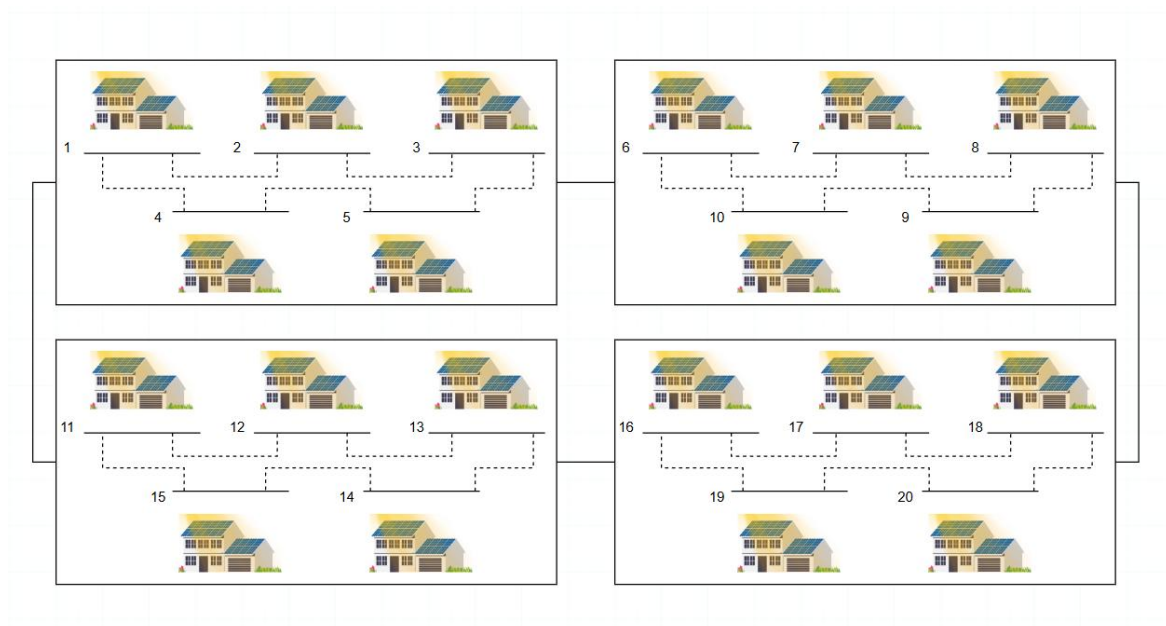


Figure 3: Test Case 3: Clustered 20-Bus System

5. Test Strategy

5.1 Unit Testing Strategy

The goal of unit testing is to verify that individual components of the DC microgrid model and optimization framework operate correctly in isolation. This includes testing functions related to system initialization, power flow calculations, objective functions, constraint handling, and battery SOC updates.

5.2 Integration Testing Strategy

The goal of integration testing is to ensure that all components of the DC microgrid simulation and OPF framework work together seamlessly. Integration points include the interaction between system data initialization, optimization routines, power flow calculations, and result visualization. The approach involves running full 24-hour simulations across different test cases, verifying consistent data flow between modules, and ensuring that outputs such as voltage profiles, power sharing, and losses are correctly computed and stored for analysis.

5.3 Validation Testing

Validation testing ensures that the system produces realistic and physically consistent results by verifying expected system behaviour under varying operating conditions, such as changes in PV availability and grid connection mode.

6. Test Cases

Test ID	Test Type	Description	Input	Expected Result	Actual Result
UT-01	Unit Test	Verify correct initialization of the 7-bus system	sys = build_tc1_7bus();	sys.N = 7, correct bus and branch data created, voltage/SOC/battery limits assigned correctly	Function executed successfully and returned a struct with N = 7, E = 6, and correctly initialized system parameters

Command Window

```
>> build_tc1_7bus|  
  
ans =  
  
    struct with fields:  
  
        N: 7  
        E: 6  
        from: [6×1 double]  
        to: [6×1 double]  
        r: [6×1 double]  
        Imax: [6×1 double]  
        v_min: 0.9025  
        v_max: 1.1025  
        PG_peak: [7×1 double]  
        PL_peak: [7×1 double]  
        has_pv: [7×1 logical]  
        has_bat: [7×1 logical]  
        SOC0: [7×1 double]  
        SOC_min: [7×1 double]  
        SOC_max: [7×1 double]  
        Cb: [7×1 double]  
        PB_min: [7×1 double]  
        PB_max: [7×1 double]  
        conv: [1×1 struct]  
        grid: [1×1 struct]  
        inc_from: [7×6 double]  
        inc_to: [7×6 double]
```

 >> |

Test ID	Test Type	Description	Input	Expected Result	Actual Result
UT-02	Unit Test	Verify correct initialization of the modified 14-bus DC system	sys = build_tc2_14bus();	sys.N = 14, Bus 14 is central load, PV/battery parameters assigned correctly	Function executed successfully and returned a struct with N = 14 and E = 20, with Bus 14 correctly configured as the central load. All system parameters were properly initialized

Command Window

```
>> build_tc2_14bus

ans =

    struct with fields:

        N: 14
        E: 20
        from: [20×1 double]
        to: [20×1 double]
        r: [20×1 double]
        Imax: [20×1 double]
        v_min: 0.9025
        v_max: 1.1025
        PG_peak: [14×1 double]
        PL_peak: [14×1 double]
        has_pv: [14×1 logical]
        has_bat: [14×1 logical]
        SOC0: [14×1 double]
        SOC_min: [14×1 double]
        SOC_max: [14×1 double]
        Cb: [14×1 double]
        PB_min: [14×1 double]
        PB_max: [14×1 double]
        conv: [1×1 struct]
        grid: [1×1 struct]
        inc_from: [14×20 double]
        inc_to: [14×20 double]
```

fx >>

Test ID	Test Type	Description	Input	Expected Result	Actual Result
UT-03	Unit Test	Verify correct initialization of the clustered 20-bus system	sys = build_tc3_20bus();	sys.N = 20, sys.E = 24, Bus 20 is central load, PV/battery parameters assigned correctly	Function executed successfully and returned a valid 20-bus system with correct initialization of parameters

Command Window

```
>> build_tc3_20bus

ans =

    struct with fields:

        N: 20
        E: 24
        from: [24x1 double]
        to: [24x1 double]
        r: [24x1 double]
        Imax: [24x1 double]
        v_min: 0.9025
        v_max: 1.1025
        PG_peak: [20x1 double]
        PL_peak: [20x1 double]
        has_pv: [20x1 logical]
        has_bat: [20x1 logical]
        SOC0: [20x1 double]
        SOC_min: [20x1 double]
        SOC_max: [20x1 double]
        Cb: [20x1 double]
        PB_min: [20x1 double]
        PB_max: [20x1 double]
        conv: [1x1 struct]
        grid: [1x1 struct]
        inc_from: [20x24 double]
        inc_to: [20x24 double]
```

fx >>

Test ID	Test Type	Description	Input	Expected Result	Actual Result
UT-04	Unit Test	Verify that case14_dc_modified.m correctly converts the IEEE 14-bus AC system into a DC-compatible structure	case14_dc_modified();	A valid MATLAB struct is returned representing a modified 14-bus DC system, with appropriate fields and dimensions consistent with the IEEE 14-bus case	Function executed successfully and returned a struct with fields, confirming correct initialization of the DC-modified system

```

Command Window

>> case14_dc_modified

ans =

    struct with fields:

    version: '2'
    baseMVA: 100
    bus: [14x13 double]
    gen: [5x21 double]
    branch: [20x13 double]
    gencost: [5x7 double]
    bus_name: {14x1 cell}

fx >>

```

Test ID	Test Type	Description	Input	Expected Result	Actual Result
UT-05	Unit Test	Verify that read_pv_and_load_csv.m correctly reads PV and load data and returns structured time-series profiles	read_pv_and_load_csv(...);	A structured output is returned containing time-series data for PV and load, all with consistent dimensions (24×1)	Function executed successfully, displaying “Successfully read load data from sheet: Test Case 1” and returned a struct with fields, confirming correct data import and formatting

Command Window

```
>> read_pv_and_load_csv
Reading load data from sheet: Test Case 1
Successfully read load data from sheet: Test Case 1

ans =

    struct with fields:

        Year: [24×1 double]
        Month: [24×1 double]
        Day: [24×1 double]
        Hour: [24×1 double]
        Minute: [24×1 double]
        DNI_Wm2: [24×1 double]
        PV_profile: [24×1 double]
        Load_profile: [24×1 double]
        Load_kWh_raw: [24×1 double]
        DateTime: [24×1 datetime]
```

fx >>

Test ID	Test Type	Description	Input	Expected Result	Actual Result
IT-01	Integration Test	Verify that run_tc1(true) executes the full TC1 workflow successfully in grid-connected mode	run_tc1(true)	The TC1 7-bus radial microgrid should initialize in grid-connected mode, solve the 24-hour OPF without execution errors	TC1 executed successfully in grid-connected mode

===== TC1 OPF2 =====

Grid connected mode: 1
Successful hours: 24/24
Total dist loss (p.u.): 0.018194
Total conv loss (p.u.): 0.244430
Total grid conv loss (p.u.): 0.048014
Total system loss (p.u.): 0.310638
Optimization runtime: 26.5543 seconds

===== TC1-7bus-grid - OPF-2 (Hour 20 snapshot) =====

Bus No.	Pi (p.u.)	PBi (p.u.)	SOCi (%)	NodalVoltage (p.u.)	Converter Efficiency (%)
1	-0.239	0	0	1.035	"97.659"
2	0.141	-0.188	85.944	1.046	"97.567"
3	0.041	-0.102	86.284	1.048	"96.150"
4	0.013	-0.05	87.869	1.049	"91.375"
5	0.032	-0.09	87.582	1.048	"95.487"
6	0.01	-0.051	87.66	1.049	"89.397"
7	0.005	-0.071	87.596	1.049	"83.159"

Test ID	Test Type	Description	Input	Expected Result	Actual Result
IT-02	Integration Test	Verify that run_tc1(false) executes the full TC1 workflow successfully in islanded mode	run_tc1(false)	The TC1 7-bus radial microgrid should initialize in islanded mode, solve the 24-hour OPF without execution errors	TC1 executed successfully in islanded mode

===== TC1 OPF2 =====

Grid connected mode: 0

Successful hours: 24/24

Total dist loss (p.u.): 0.018168

Total conv loss (p.u.): 0.244426

Total system loss (p.u.): 0.262594

Optimization runtime: 26.0157 seconds

|

===== TC1-7bus-islanded - OPF-2 (Hour 20 snapshot) =====

Bus No.	Pi (p.u.)	PBi (p.u.)	SOCi (%)	NodalVoltage (p.u.)	Converter Efficiency (%)
1	-0.239	0	0	1.036	"97.660"
2	0.143	-0.187	83.668	1.047	"97.573"
3	0.042	-0.102	86.535	1.049	"96.202"
4	0.011	-0.047	87.304	1.05	"90.586"
5	0.033	-0.089	86.968	1.049	"95.621"
6	0.009	-0.052	87.653	1.05	"88.844"
7	0.003	-0.068	87.309	1.05	"75.494"

Test ID	Test Type	Description	Input	Expected Result	Actual Result
IT-03	Integration Test	Verify that run_tc2(true) executes the full TC2 workflow successfully in grid-connected mode	run_tc2(true)	The TC2 14-bus DC microgrid should initialize in grid-connected mode, solve the 24-hour OPF without execution errors	TC2 executed successfully in grid-connected mode

===== TC2 OPF2 =====

Grid connected mode: 1
Successful hours: 24/24
Total dist loss (p.u.): 0.036651
Total conv loss (p.u.): 0.500959
Total grid conv loss (p.u.): 0.048014
Total system loss (p.u.): 0.585624
Optimization runtime: 80.7177 seconds

===== TC2-14bus-grid - OPF-2 (Hour 20 snapshot) =====

Bus No.	Pi (p.u.)	PBi (p.u.)	SOCi (%)	NodalVoltage (p.u.)	Converter Efficiency (%)
1	0.026	-0.024	82.108	1.049	"94.902"
2	0.047	-0.202	81.124	1.048	"96.429"
3	-0.172	-0.4	74.477	1.045	"97.631"
4	0.031	-0.345	77.356	1.049	"95.469"
5	0.022	-0.076	75.982	1.049	"94.253"
6	0.018	-0.097	72.691	1.049	"93.499"
7	0.027	-0.025	84.056	1.049	"94.965"
8	0.023	-0.021	79.359	1.049	"94.393"
9	0.032	-0.233	81.502	1.049	"95.485"
10	0.013	-0.076	85.65	1.049	"91.398"
11	0.007	-0.032	75.677	1.049	"86.761"
12	0.007	-0.05	78.892	1.049	"86.885"
13	0.038	-0.134	85.036	1.049	"95.954"
14	-0.119	0	0	1.042	"97.485"

Test ID	Test Type	Description	Input	Expected Result	Actual Result
IT-04	Integration Test	Verify that run_tc2(false) executes the full TC2 workflow successfully in islanded mode	run_tc2(false)	The TC1 14-bus DC microgrid should initialize in islanded mode, solve the 24-hour OPF without execution errors	TC2 executed successfully in islanded mode

===== TC2 OPF2 =====

Grid connected mode: 0
Successful hours: 24/24
Total dist loss (p.u.): 0.040081
Total conv loss (p.u.): 0.508212
Total system loss (p.u.): 0.548293
Optimization runtime: 72.1294 seconds

===== TC2-14bus-islanded - OPF-2 (Hour 20 snapshot) =====

Bus No.	Pi (p.u.)	PBi (p.u.)	SOCi (%)	NodalVoltage (p.u.)	Converter Efficiency (%)
1	0.026	-0.024	83.72	1.049	"94.914"
2	0.047	-0.199	81.387	1.048	"96.425"
3	-0.172	-0.4	72.371	1.045	"97.631"
4	0.032	-0.349	76.646	1.049	"95.505"
5	0.022	-0.075	77.88	1.049	"94.169"
6	0.018	-0.097	83.767	1.049	"93.473"
7	0.027	-0.025	83.921	1.049	"94.996"
8	0.022	-0.02	87.187	1.049	"94.300"
9	0.032	-0.234	80.554	1.049	"95.495"
10	0.013	-0.077	77.351	1.049	"91.704"
11	0.007	-0.032	81.897	1.049	"86.505"
12	0.007	-0.05	82.695	1.049	"86.801"
13	0.038	-0.133	78.359	1.049	"95.951"
14	-0.119	0	0	1.042	"97.485"

Test ID	Test Type	Description	Input	Expected Result	Actual Result
IT-05	Integration Test	Verify that run_tc3(true) executes the full TC3 workflow successfully in grid-connected mode	run_tc3(true)	The TC3 20-bus clustered microgrid should initialize in grid-connected mode, solve the 24-hour OPF without execution errors	TC3 executed successfully in grid-connected mode

===== OPF2 =====

Grid connected mode: 1
Successful hours: 24/24
Total dist loss (p.u.): 0.024041
Total conv loss (p.u.): 0.616251
Total grid conv loss (p.u.): 0.048014
Total system loss (p.u.): 0.688306
Optimization runtime: 149.3505 seconds

===== TC3-20bus-grid - OPF-2 (Hour 20 snapshot) =====

Bus No.	Pi (p.u.)	PBi (p.u.)	SOCi (%)	NodalVoltage (p.u.)	Converter Efficiency (%)
1	0.002	-0.097	87.004	1.047	"65.988"
2	0.002	-0.048	87.634	1.047	"67.772"
3	0.003	-0.097	87.043	1.047	"71.799"
4	0.002	-0.041	88.426	1.047	"67.874"
5	0.002	-0.049	87.75	1.047	"66.007"
6	0	-0.061	87.647	1.047	"Non-Op"
7	0	-0.058	87.915	1.047	"Non-Op"
8	0	-0.06	87.424	1.047	"Non-Op"
9	0	-0.087	86.433	1.047	"Non-Op"
10	0	-0.074	86.567	1.047	"Non-Op"
11	0.002	-0.049	87.355	1.047	"67.012"
12	0.002	-0.092	86.926	1.047	"69.238"
13	0.003	-0.049	88.05	1.047	"73.811"
14	0.002	-0.049	88.123	1.047	"68.788"
15	0.002	-0.1	85.542	1.047	"66.573"
16	0.147	-0.201	83.62	1.044	"97.584"
17	0.046	-0.093	87.25	1.046	"96.353"
18	0.03	-0.116	86.669	1.046	"95.300"
19	0.148	-0.233	80.447	1.044	"97.587"
20	-0.399	0	0	1.034	"97.515"

Test ID	Test Type	Description	Input	Expected Result	Actual Result
IT-06	Integration Test	Verify that run_tc3(false) executes the full TC3 workflow successfully in islanded mode	run_tc3(false)	The TC3 20-bus clustered microgrid should initialize in islanded mode, solve the 24-hour OPF without execution errors	TC3 executed successfully in islanded mode

```

===== OPF2 =====
Grid connected mode: 0
Successful hours: 24/24
Total dist loss (p.u.): 0.023932
Total conv loss (p.u.): 0.616246
Total system loss (p.u.): 0.640179
Optimization runtime: 130.6763 seconds

```

```

===== TC3-20bus-islanded - OPF-2 (Hour 20 snapshot) =====

```

Bus No.	Pi (p.u.)	PBi (p.u.)	SOCi (%)	NodalVoltage (p.u.)	Converter Efficiency (%)
1	0	-0.096	86.659	1.049	"Non-Op"
2	0	-0.047	87.68	1.049	"Non-Op"
3	0.002	-0.095	86.143	1.049	"70.640"
4	0	-0.041	88.455	1.049	"Non-Op"
5	0	-0.049	87.793	1.049	"Non-Op"
6	0	-0.06	87.602	1.05	"Non-Op"
7	0	-0.057	87.86	1.05	"Non-Op"
8	0	-0.059	87.4	1.049	"Non-Op"
9	0	-0.085	86.487	1.05	"Non-Op"
10	0	-0.074	86.701	1.05	"Non-Op"
11	0	-0.047	87.272	1.049	"Non-Op"
12	0	-0.091	87.008	1.049	"Non-Op"
13	0.003	-0.049	88.109	1.049	"73.475"
14	0	-0.048	88.047	1.049	"Non-Op"
15	0	-0.096	85.266	1.049	"Non-Op"
16	0.148	-0.205	83.668	1.046	"97.588"
17	0.048	-0.091	87.192	1.048	"96.433"
18	0.035	-0.125	86.41	1.049	"95.736"
19	0.15	-0.237	81.301	1.046	"97.593"
20	-0.399	0	0	1.037	"97.515"

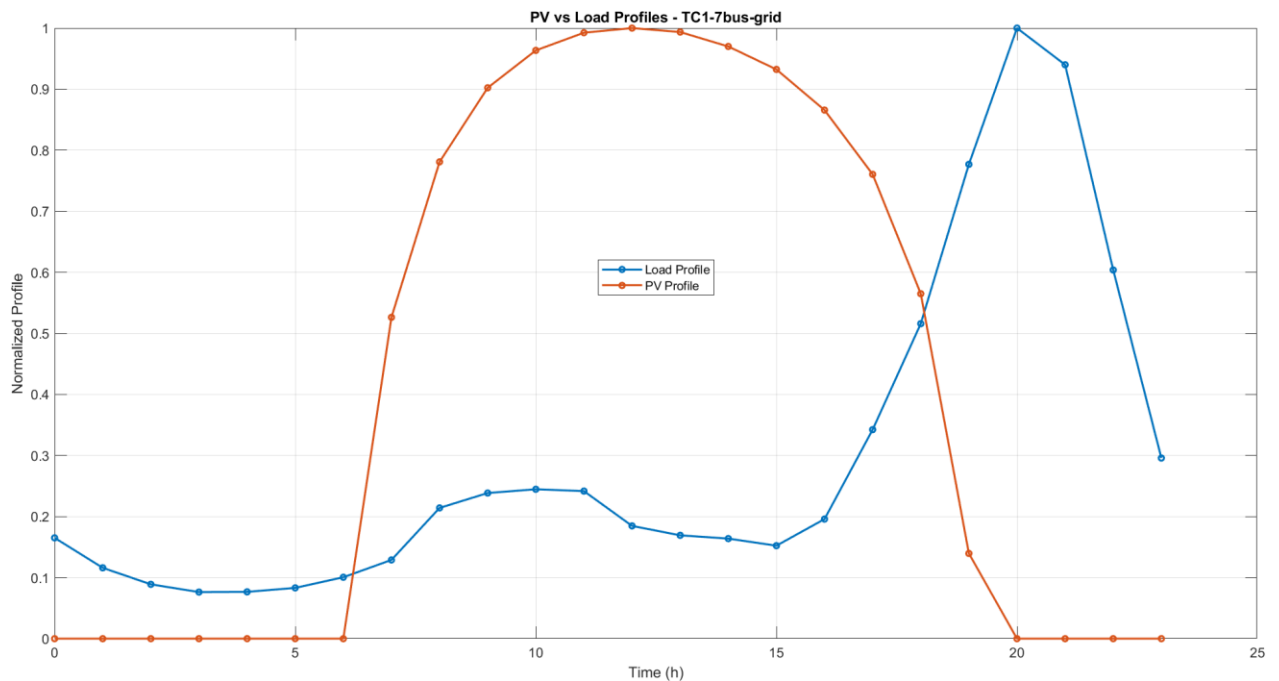
Test ID	Test Type	Description	Input	Expected Result	Actual Result
IT-07	Integration Test	Verify that TC1 runs correctly using CSV-based PV and load profiles	run_tc1(true) with valid CSV files present	The system should successfully read PV and load data from the specified CSV and Excel files and generate normalized profiles	TC1 executed successfully using CSV-based profiles. The system read the PV and load data correctly and generated valid 24-hour profiles

```

Reading load data from sheet: Test Case 1
Successfully read load data from sheet: Test Case 1

===== TC1 OPF2 =====
Grid connected mode: 1
PV availability scale: 1.00
Successful hours: 24/24
Total dist loss (p.u.): 0.018194
Total conv loss (p.u.): 0.244430
Total grid conv loss (p.u.): 0.048014
Total system loss (p.u.): 0.310638
Optimization runtime: 60.6357 seconds

```



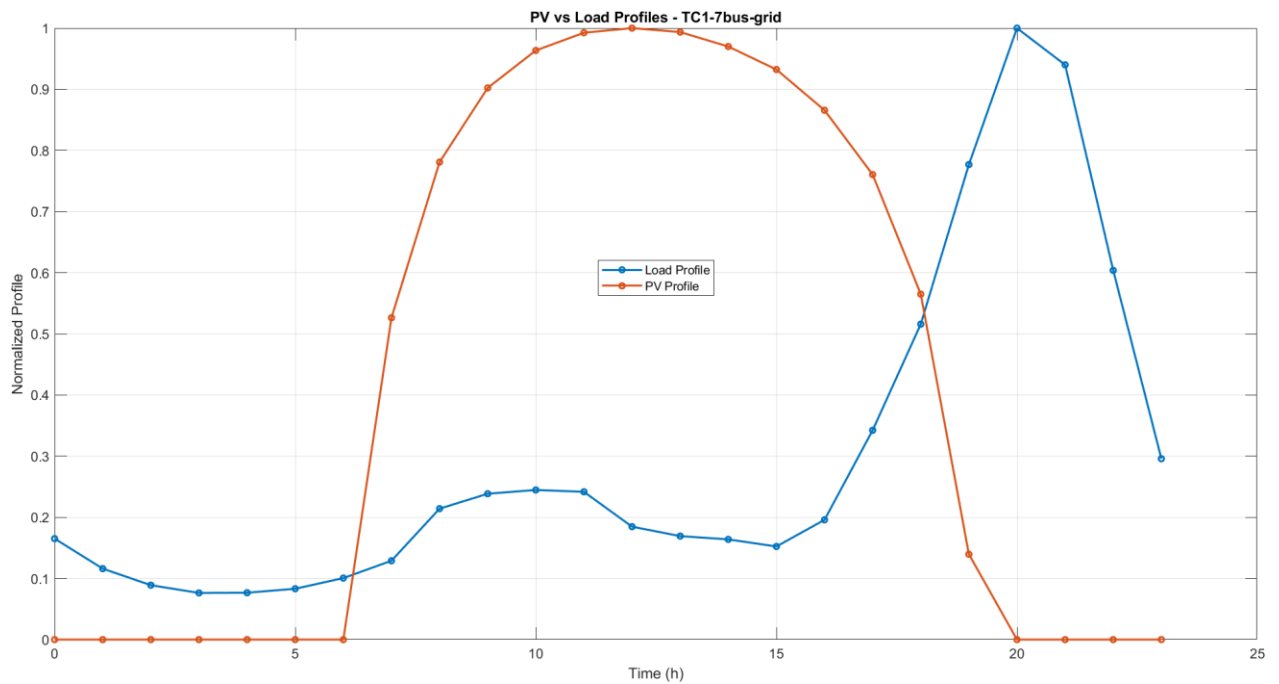
Test ID	Test Type	Description	Input	Expected Result	Actual Result
IT-08	Integration Test	Verify that TC2 runs correctly using CSV-based PV and load profiles	run_tc2(true) with valid CSV files present	The system should successfully read PV and load data from the specified CSV and Excel files and generate normalized profiles	TC2 executed successfully using CSV-based profiles. The system read the PV and load data correctly and generated valid 24-hour profiles

```

Reading load data from sheet: Test Case 2
Successfully read load data from sheet: Test Case 2

===== TC2 OPF2 =====
Grid connected mode: 1
PV availability scale: 1.00
Successful hours: 24/24
Total dist loss (p.u.): 0.037284
Total conv loss (p.u.): 0.500366
Total grid conv loss (p.u.): 0.048018
Total system loss (p.u.): 0.585669
Optimization runtime: 181.0294 seconds

```



Test ID	Test Type	Description	Input	Expected Result	Actual Result
IT-09	Integration Test	Verify that TC3 runs correctly using CSV-based PV and load profiles	run_tc3(true) with valid CSV files present	The system should successfully read PV and load data from the specified CSV and Excel files and generate normalized profiles	TC3 executed successfully using CSV-based profiles. The system read the PV and load data correctly and generated valid 24-hour profiles

```

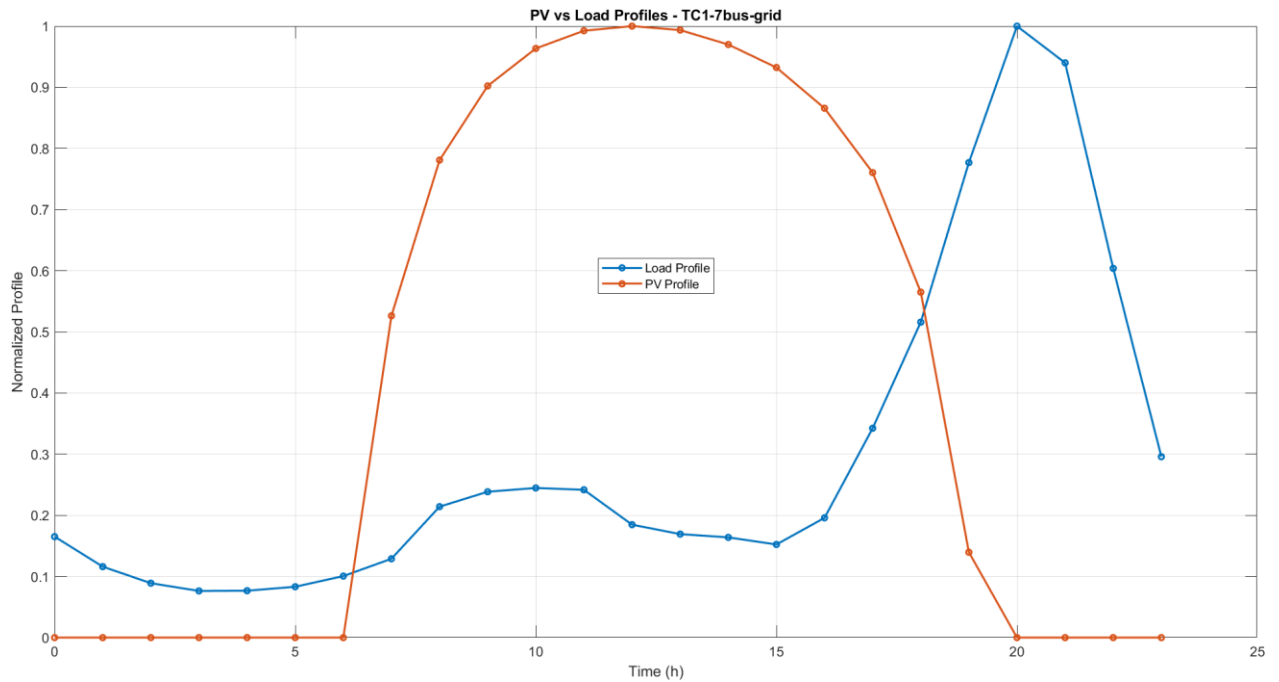
Reading load data from sheet: Test Case 3
Successfully read load data from sheet: Test Case 3

```

```

===== OPF2 =====
Grid connected mode: 1
PV availability scale: 1.00
Successful hours: 24/24
Total dist loss (p.u.): 0.024041
Total conv loss (p.u.): 0.616251
Total grid conv loss (p.u.): 0.048014
Total system loss (p.u.): 0.688306
Optimization runtime: 330.3539 seconds

```



Test ID	Test Type	Description	Input	Expected Result	Actual Result
IT-10	Integration Test	Verify that TC1 runs correctly when PV availability is set to full generation	run_tc1(true) with pv_scale = 1.0	The system should use the full imported PV profile without scaling reduction and complete the 24-hour OPF simulation successfully	TC1 executed successfully with pv_scale = 1.0

Command Window

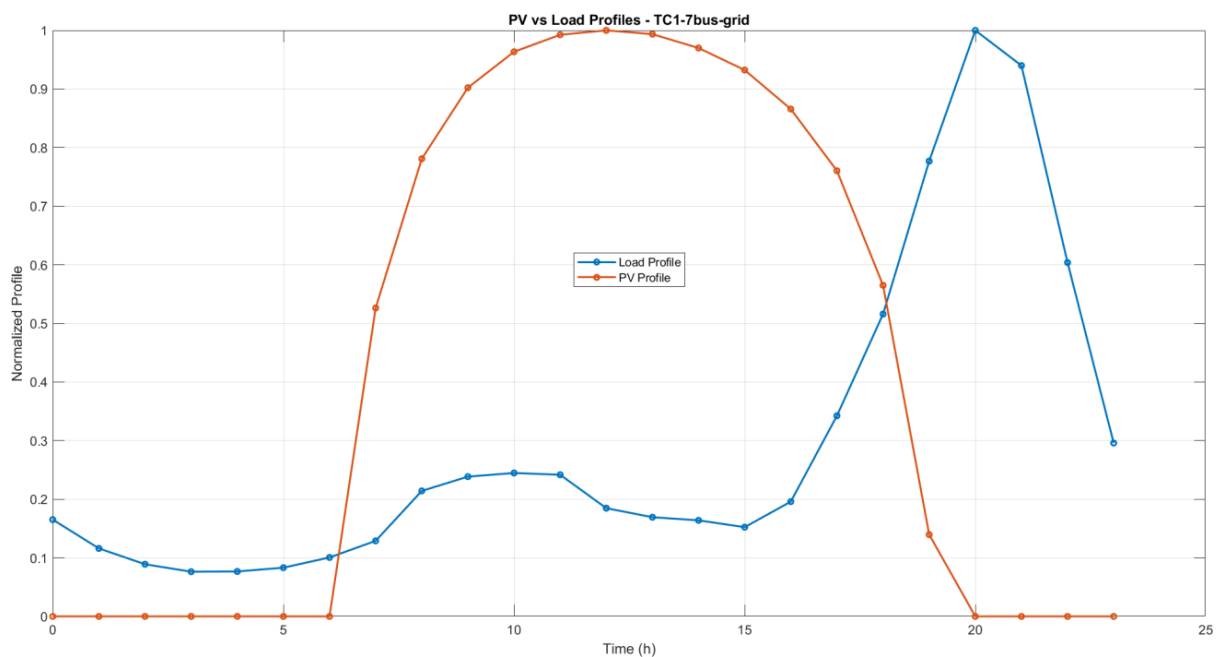
```

Set PV availability level for TC1 (0 to 1):
  1 = Full PV generation
  0 = No PV generation
Enter PV availability (0-1): 1
Reading load data from sheet: Test Case 1
Successfully read load data from sheet: Test Case 1

===== TC1 OPF2 =====

Grid connected mode: 1
PV availability scale: 1.00
Successful hours: 24/24
Total dist loss (p.u.): 0.018194
Total conv loss (p.u.): 0.244430
Total grid conv loss (p.u.): 0.048014
Total system loss (p.u.): 0.310638
Optimization runtime: 57.2978 seconds

```



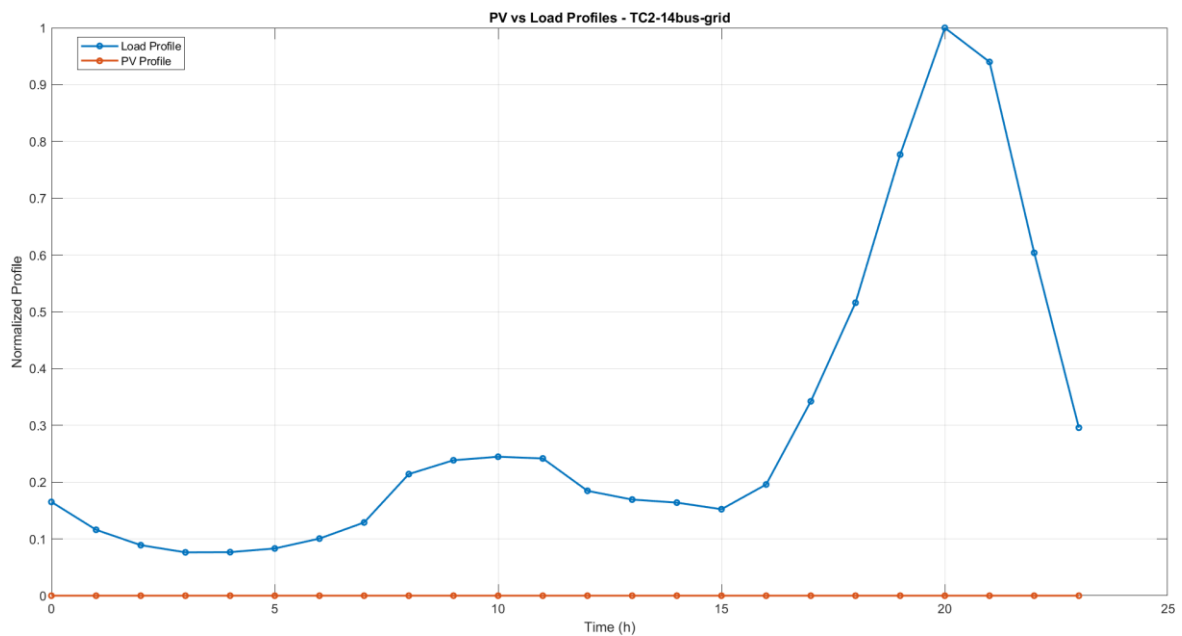
Test ID	Test Type	Description	Input	Expected Result	Actual Result
IT-11	Integration Test	Verify that TC1 runs correctly when PV availability is set to zero	run_tc1(true) with pv_scale = 0.0	The PV profile should be reduced to zero for all 24 hours, and the system should still solve successfully using battery and/or grid support where applicable	TC1 executed successfully with pv_scale = 0.0

```

Set PV availability level for TC1 (0 to 1):
  1 = Full PV generation
  0 = No PV generation
Enter PV availability (0-1): 0
Reading load data from sheet: Test Case 1
Successfully read load data from sheet: Test Case 1

===== TC1 OPF2 =====
Grid connected mode: 1
PV availability scale: 0.00
Successful hours: 24/24
Total dist loss (p.u.): 0.027402
Total conv loss (p.u.): 0.260920
Total grid conv loss (p.u.): 0.069183
Total system loss (p.u.): 0.357504
Optimization runtime: 58.8639 seconds

```



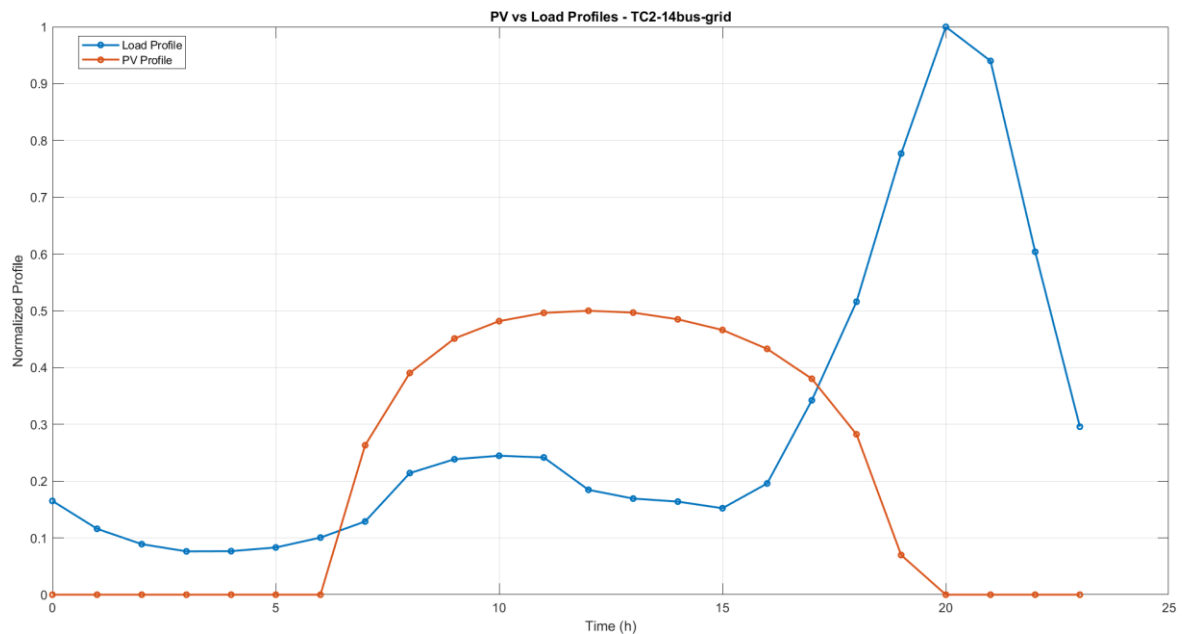
Test ID	Test Type	Description	Input	Expected Result	Actual Result
IT-12	Integration Test	Verify that TC1 runs correctly when PV availability is partially reduced	run_tc1(true) with pv_scale = 0.5	The imported PV profile should be reduced to 50% of its original magnitude, and the system should complete the OPF simulation successfully	TC1 executed successfully with pv_scale = 0.5

```

Set PV availability level for TC1 (0 to 1):
  1 = Full PV generation
  0 = No PV generation
Enter PV availability (0-1): 0.5
Reading load data from sheet: Test Case 1
Successfully read load data from sheet: Test Case 1

===== TC1 OPF2 =====
Grid connected mode: 1
PV availability scale: 0.50
Successful hours: 24/24
Total dist loss (p.u.): 0.018168
Total conv loss (p.u.): 0.244427
Total grid conv loss (p.u.): 0.048012
Total system loss (p.u.): 0.310607
Optimization runtime: 59.7288 seconds

```



Test ID	Test Type	Description	Input	Expected Result	Actual Result
IT-13	Integration Test	Verify that TC1 runs correctly when PV availability is set to a low level	run_tc1(true) with pv_scale = 0.2	The imported PV profile should be reduced to 20% of its original value, and the system should still solve successfully	TC1 executed successfully with pv_scale = 0.2

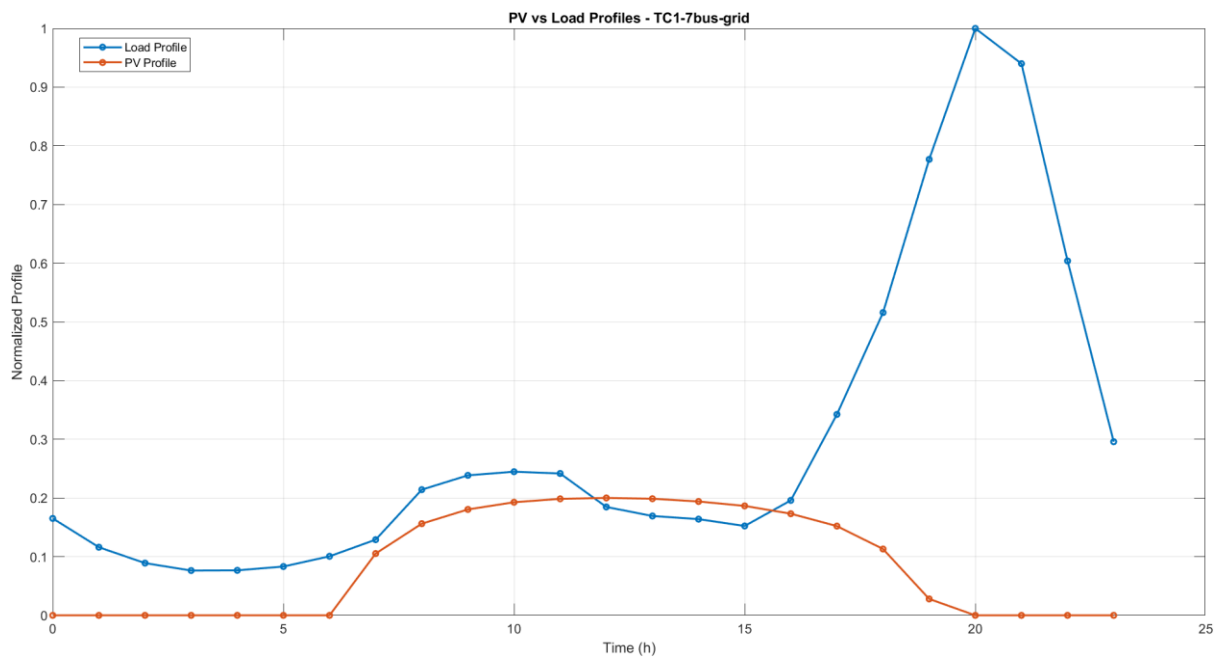
Command Window

```

Set PV availability level for TC1 (0 to 1):
  1 = Full PV generation
  0 = No PV generation
Enter PV availability (0-1): 0.2
Reading load data from sheet: Test Case 1
Successfully read load data from sheet: Test Case 1

===== TC1 OPF2 =====
Grid connected mode: 1
PV availability scale: 0.20
Successful hours: 24/24
Total dist loss (p.u.): 0.019481
Total conv loss (p.u.): 0.244125
Total grid conv loss (p.u.): 0.050032
Total system loss (p.u.): 0.313638
Optimization runtime: 60.6034 seconds

```



Test ID	Test Type	Description	Input	Expected Result	Actual Result
IT-14	Integration Test	Verify that TC2 runs correctly when PV availability is set to full generation	run_tc2(true) with pv_scale = 1.0	The system should use the full imported PV profile without scaling reduction and complete the 24-hour OPF simulation successfully	TC2 executed successfully with pv_scale = 1.0

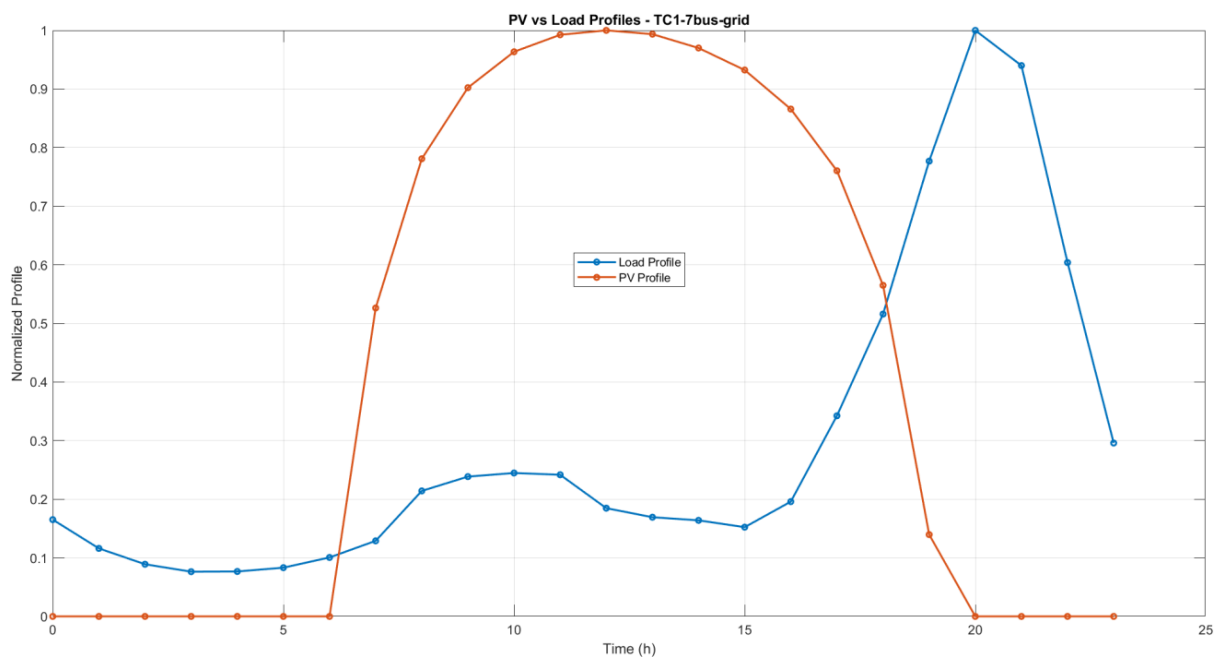
Command Window

```

Set PV availability level for TC2 (0 to 1):
  1 = Full PV generation
  0 = No PV generation
Enter PV availability (0-1): 1
Reading load data from sheet: Test Case 2
Successfully read load data from sheet: Test Case 2

===== TC2 OPF2 =====
Grid connected mode: 1
PV availability scale: 1.00
Successful hours: 24/24
Total dist loss (p.u.): 0.040620
Total conv loss (p.u.): 0.508639
Total grid conv loss (p.u.): 0.048014
Total system loss (p.u.): 0.597274
Optimization runtime: 188.7551 seconds

```



Test ID	Test Type	Description	Input	Expected Result	Actual Result
IT-15	Integration Test	Verify that TC2 runs correctly when PV availability is set to zero	run_tc2(true) with pv_scale = 0.0	The PV profile should be reduced to zero for all 24 hours, and the system should still solve successfully using battery and/or grid support where applicable	TC2 executed successfully with pv_scale = 0.0

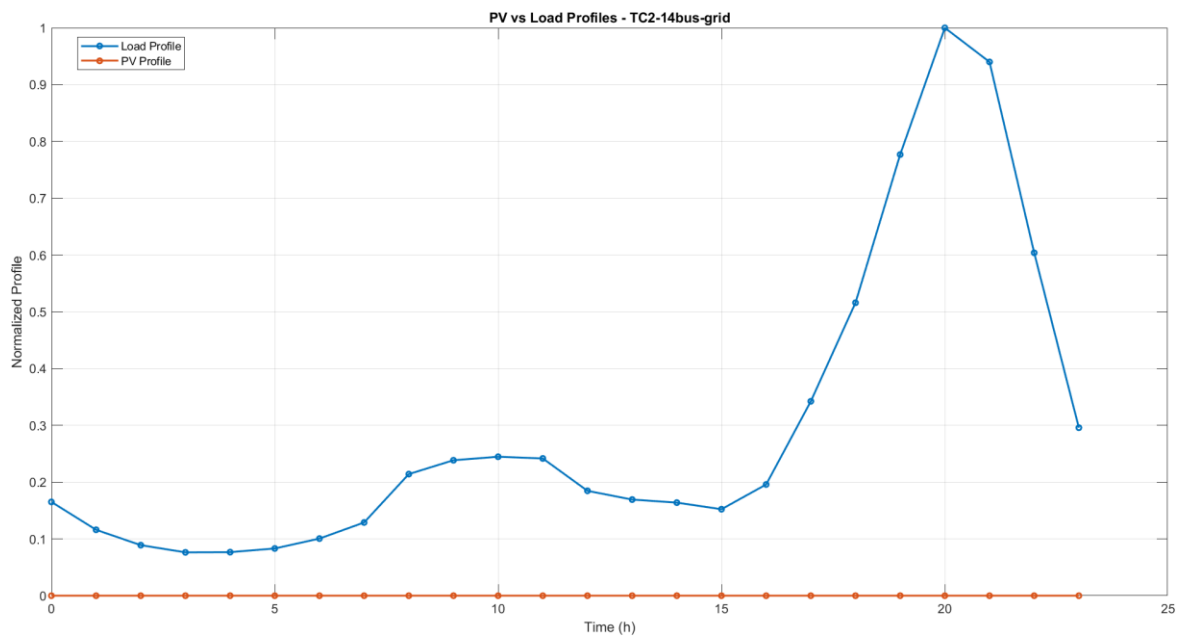
Command Window

```

Set PV availability level for TC2 (0 to 1):
  1 = Full PV generation
  0 = No PV generation
Enter PV availability (0-1): 0
Reading load data from sheet: Test Case 2
Successfully read load data from sheet: Test Case 2

===== TC2 OPF2 =====
Grid connected mode: 1
PV availability scale: 0.00
Successful hours: 24/24
Total dist loss (p.u.): 0.250442
Total conv loss (p.u.): 1.052727
Total grid conv loss (p.u.): 0.460681
Total system loss (p.u.): 1.763850
Optimization runtime: 74.7637 seconds

```



Test ID	Test Type	Description	Input	Expected Result	Actual Result
IT-16	Integration Test	Verify that TC2 runs correctly when PV availability is partially reduced	run_tc2(true) with pv_scale = 0.5	The imported PV profile should be reduced to 50% of its original magnitude, and the system should complete the OPF simulation successfully	TC2 executed successfully with pv_scale = 0.5

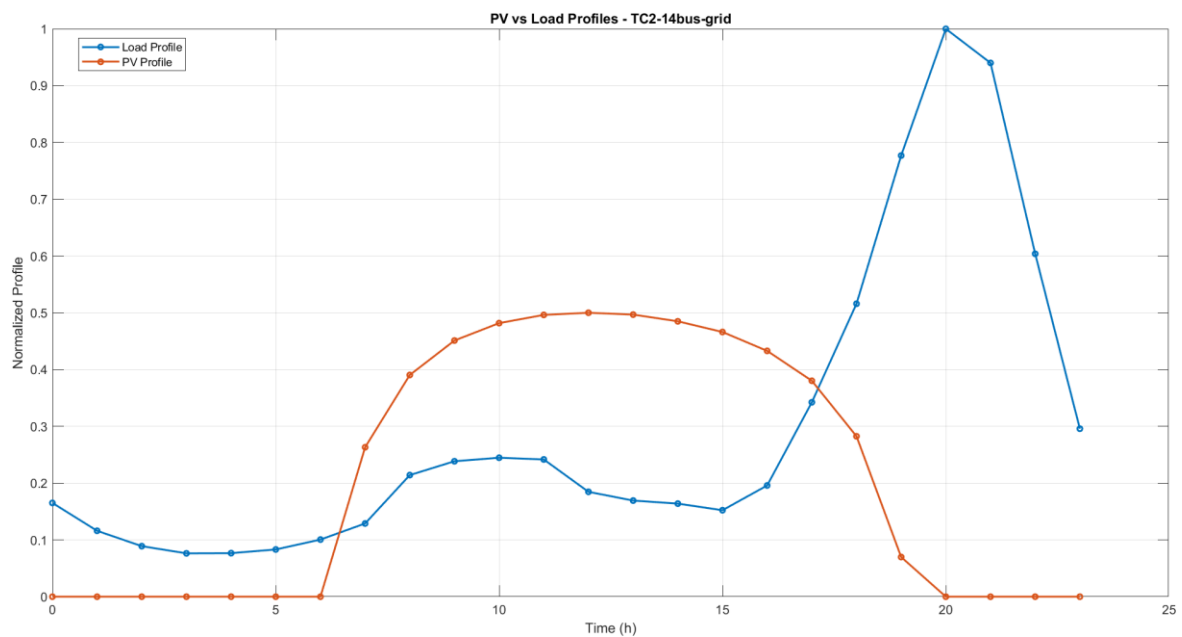
Command Window

```

Set PV availability level for TC2 (0 to 1):
  1 = Full PV generation
  0 = No PV generation
Enter PV availability (0-1): 0.5
Reading load data from sheet: Test Case 2
Successfully read load data from sheet: Test Case 2

===== TC2 OPF2 =====
Grid connected mode: 1
PV availability scale: 0.50
Successful hours: 24/24
Total dist loss (p.u.): 0.067461
Total conv loss (p.u.): 0.579183
Total grid conv loss (p.u.): 0.048014
Total system loss (p.u.): 0.694658
Optimization runtime: 80.3222 seconds

```



Test ID	Test Type	Description	Input	Expected Result	Actual Result
IT-17	Integration Test	Verify that TC2 runs correctly when PV availability is set to a low level	run_tc2(true) with pv_scale = 0.2	The imported PV profile should be reduced to 20% of its original value, and the system should still solve successfully	TC2 executed successfully with pv_scale = 0.2

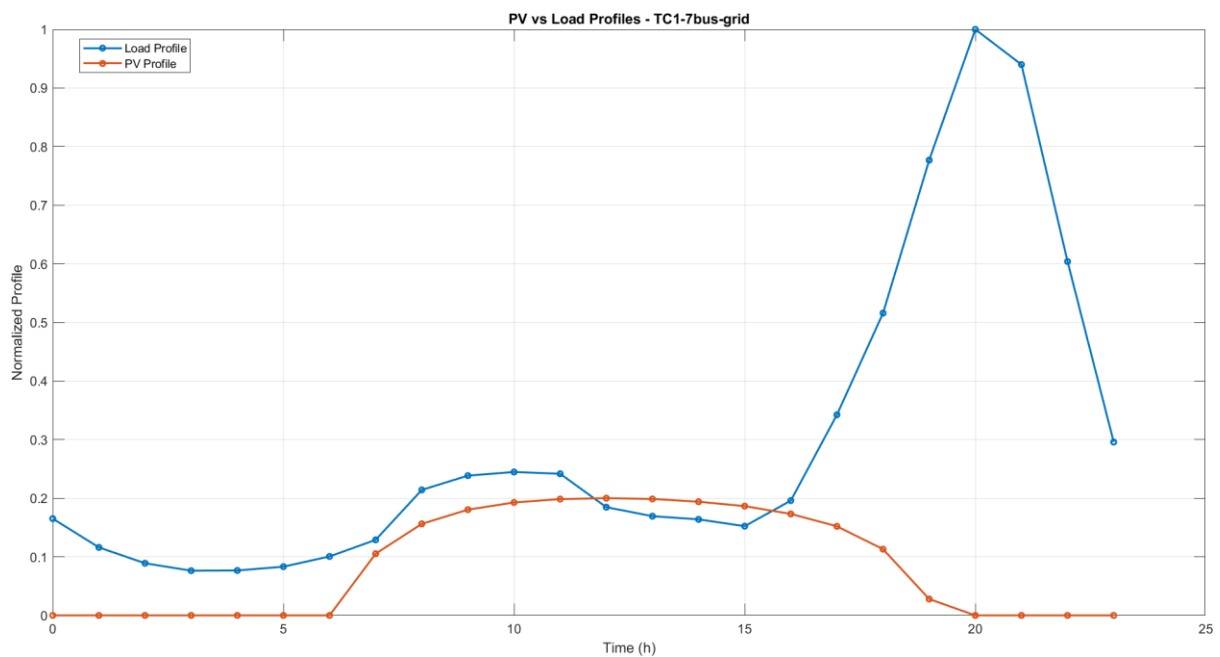
Command Window

```

Set PV availability level for TC2 (0 to 1):
  1 = Full PV generation
  0 = No PV generation
Enter PV availability (0-1): 0.2
Reading load data from sheet: Test Case 2
Successfully read load data from sheet: Test Case 2

===== TC2 OPF2 =====
Grid connected mode: 1
PV availability scale: 0.20
Successful hours: 24/24
Total dist loss (p.u.): 0.126773
Total conv loss (p.u.): 0.732537
Total grid conv loss (p.u.): 0.108326
Total system loss (p.u.): 0.967636
Optimization runtime: 76.3174 seconds

```



Test ID	Test Type	Description	Input	Expected Result	Actual Result
IT-18	Integration Test	Verify that TC3 runs correctly when PV availability is set to full generation	run_tc3(true) with pv_scale = 1.0	The system should use the full imported PV profile without scaling reduction and complete the 24-hour OPF simulation successfully	TC3 executed successfully with pv_scale = 1.0

```

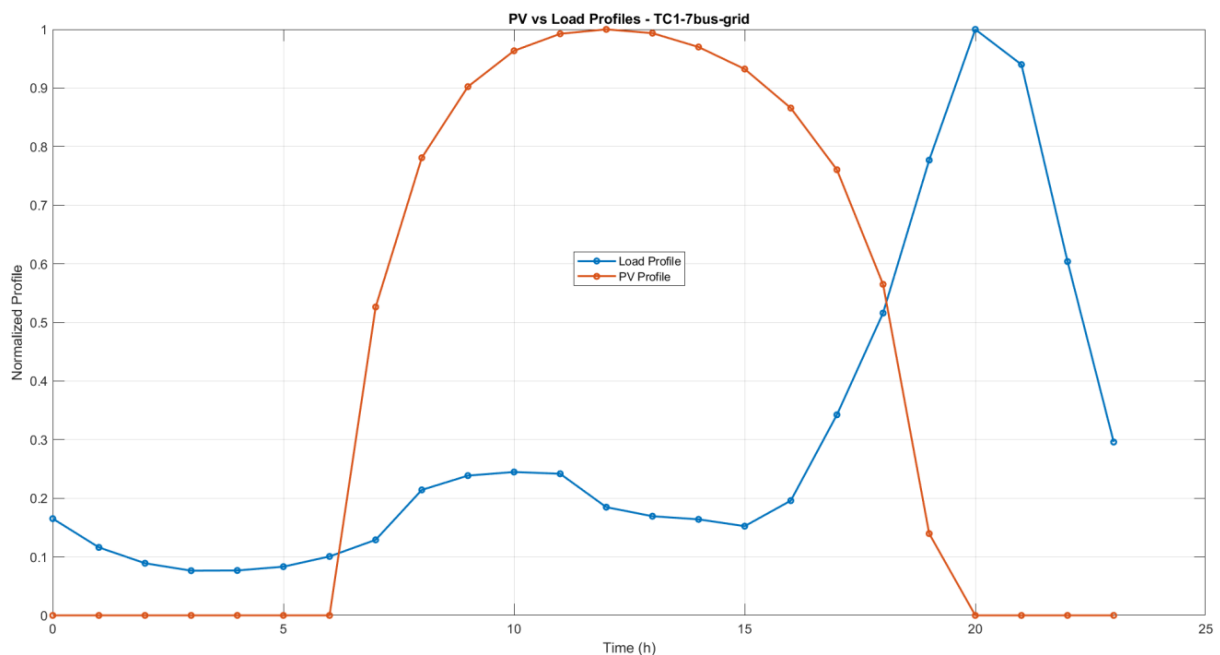
Command Window

Select grid connection mode for TC3:
  1 = Grid-connected
  0 = Islanded
Enter choice (1 or 0): 1

Set PV availability level for TC3 (0 to 1):
  1 = Full PV generation
  0 = No PV generation
Enter PV availability (0-1): 1
Reading load data from sheet: Test Case 3
Successfully read load data from sheet: Test Case 3

===== OPF2 =====
Grid connected mode: 1
PV availability scale: 1.00
Successful hours: 24/24
Total dist loss (p.u.): 0.024041
Total conv loss (p.u.): 0.616251
Total grid conv loss (p.u.): 0.048014
Total system loss (p.u.): 0.688306
Optimization runtime: 173.7287 seconds

```



Test ID	Test Type	Description	Input	Expected Result	Actual Result
IT-19	Integration Test	Verify that TC3 runs correctly when PV availability is set to zero	run_tc3(true) with pv_scale = 0.0	The PV profile should be reduced to zero for all 24 hours, and the system should still solve successfully using battery and/or grid support where applicable	TC3 executed successfully with pv_scale = 0.0

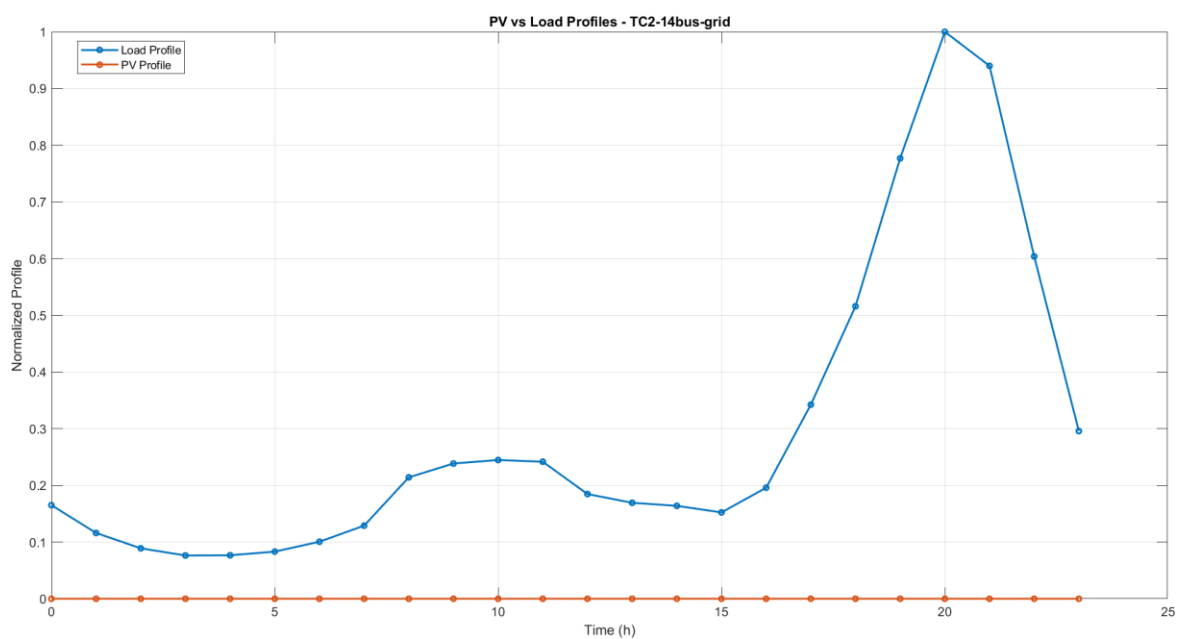
Command Window

```

Set PV availability level for TC3 (0 to 1):
  1 = Full PV generation
  0 = No PV generation
Enter PV availability (0-1): 0
Reading load data from sheet: Test Case 3
Successfully read load data from sheet: Test Case 3

===== OPF2 =====
Grid connected mode: 1
PV availability scale: 0.00
Successful hours: 24/24
Total dist loss (p.u.): 0.148074
Total conv loss (p.u.): 0.697362
Total grid conv loss (p.u.): 0.051117
Total system loss (p.u.): 0.896553
Optimization runtime: 123.5511 seconds

```



Test ID	Test Type	Description	Input	Expected Result	Actual Result
IT-20	Integration Test	Verify that TC3 runs correctly when PV availability is partially reduced	run_tc3(true) with pv_scale = 0.5	The imported PV profile should be reduced to 50% of its original magnitude, and the system should complete the OPF simulation successfully	TC3 executed successfully with pv_scale = 0.5

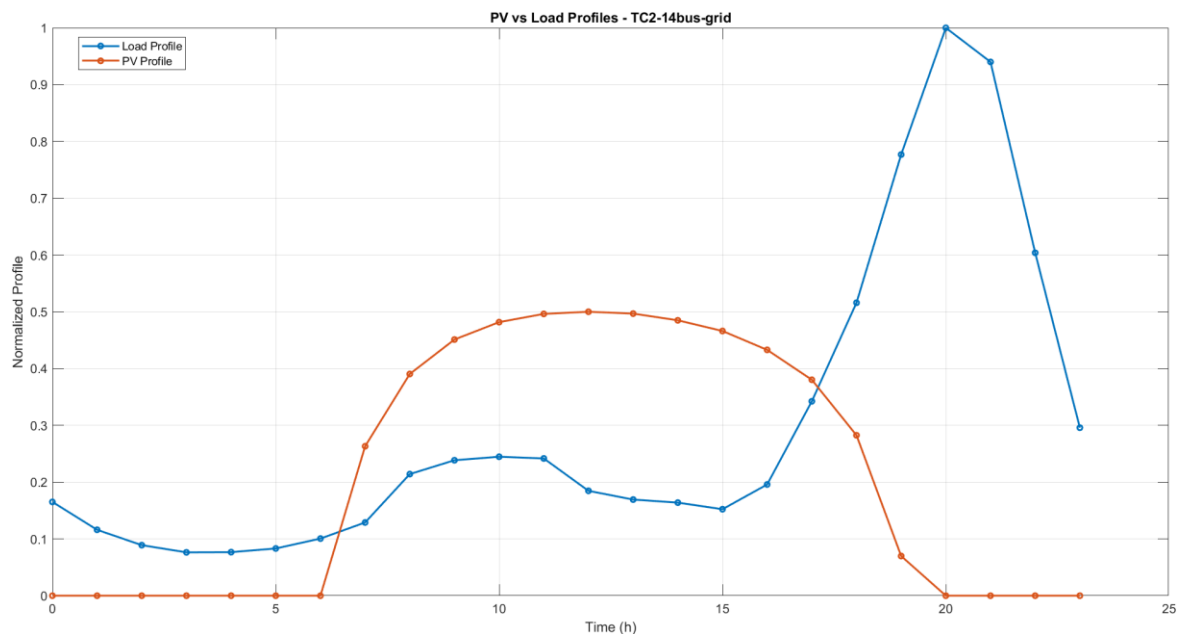
Command Window

```

Set PV availability level for TC3 (0 to 1):
  1 = Full PV generation
  0 = No PV generation
Enter PV availability (0-1): 0.5
Reading load data from sheet: Test Case 3
Successfully read load data from sheet: Test Case 3

===== OPF2 =====
Grid connected mode: 1
PV availability scale: 0.50
Successful hours: 24/24
Total dist loss (p.u.): 0.026556
Total conv loss (p.u.): 0.621032
Total grid conv loss (p.u.): 0.048012
Total system loss (p.u.): 0.695600
Optimization runtime: 157.6453 seconds

```



Test ID	Test Type	Description	Input	Expected Result	Actual Result
IT-21	Integration Test	Verify that TC3 runs correctly when PV availability is set to a low level	run_tc3(true) with pv_scale = 0.2	The imported PV profile should be reduced to 20% of its original value, and the system should still solve successfully	TC3 executed successfully with pv_scale = 0.2

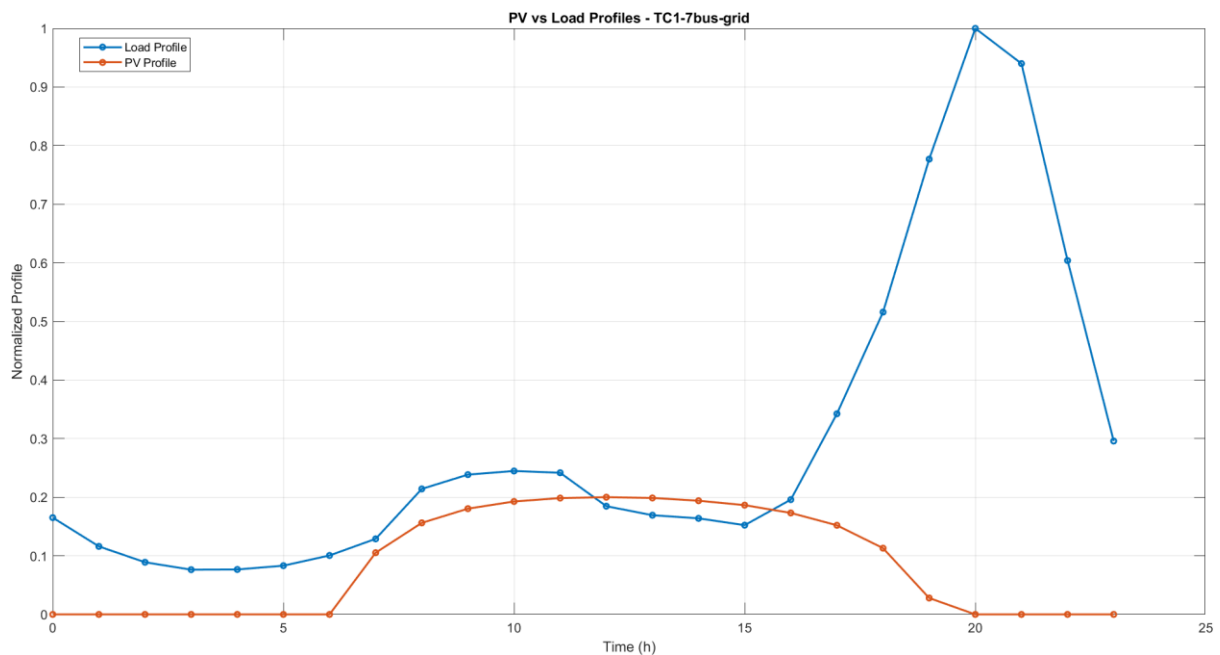
Command Window

```

Set PV availability level for TC3 (0 to 1):
    1 = Full PV generation
    0 = No PV generation
Enter PV availability (0-1): 0.2
Reading load data from sheet: Test Case 3
Successfully read load data from sheet: Test Case 3

===== OPF2 =====
Grid connected mode: 1
PV availability scale: 0.20
Successful hours: 24/24
Total dist loss (p.u.): 0.050326
Total conv loss (p.u.): 0.642646
Total grid conv loss (p.u.): 0.048014
Total system loss (p.u.): 0.740986
Optimization runtime: 158.6886 seconds

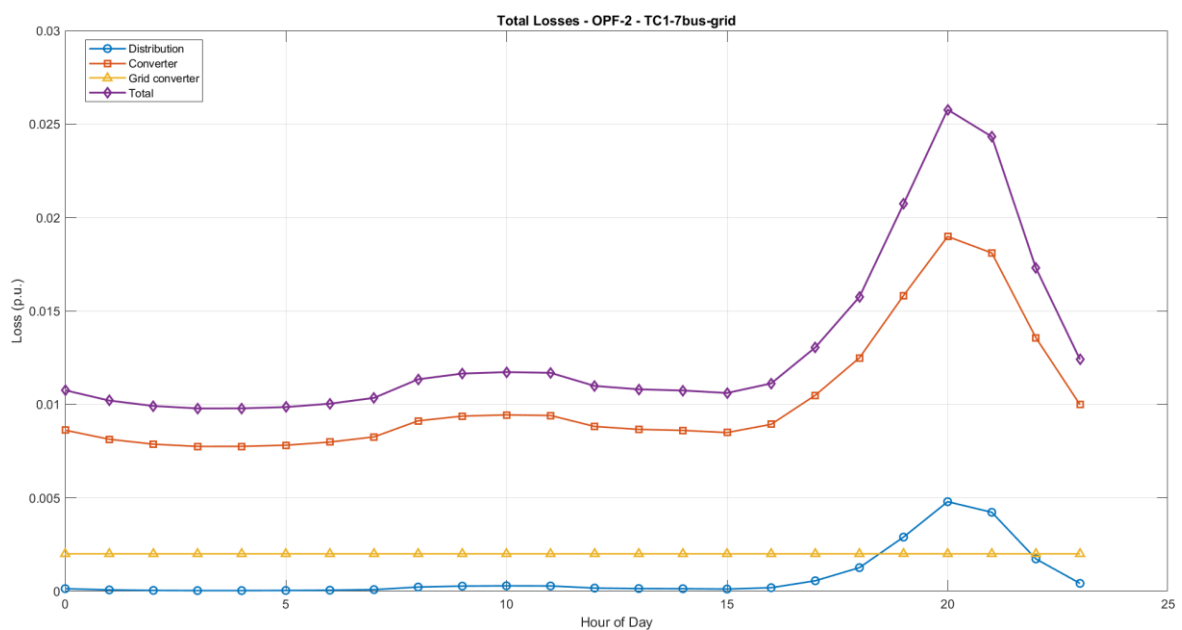
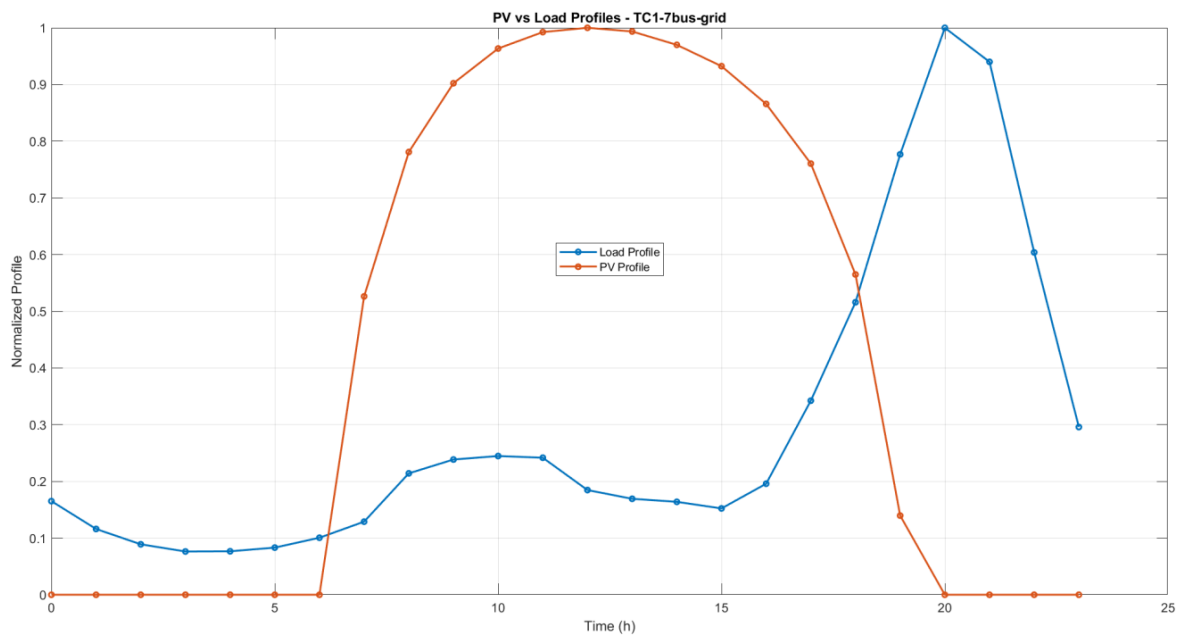
```

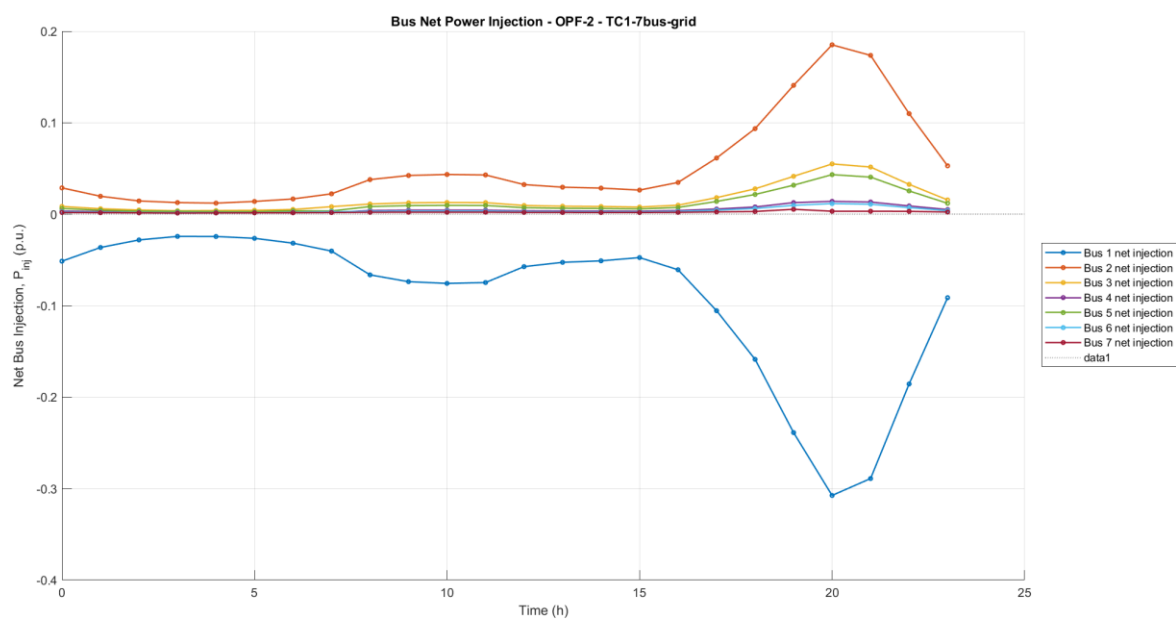
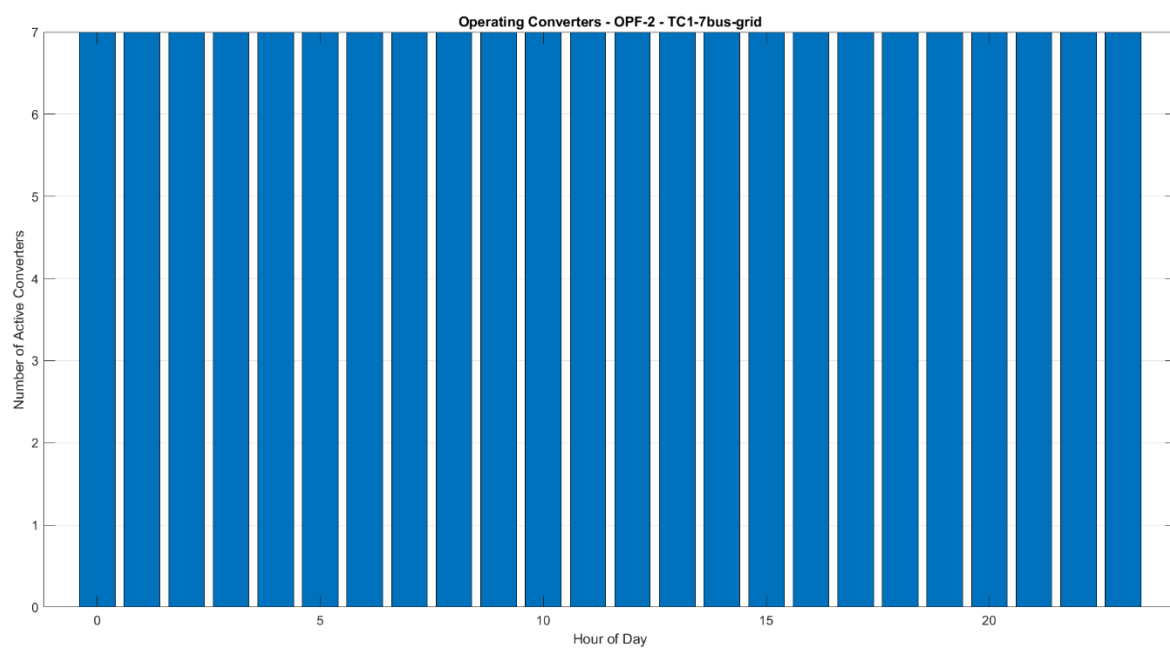


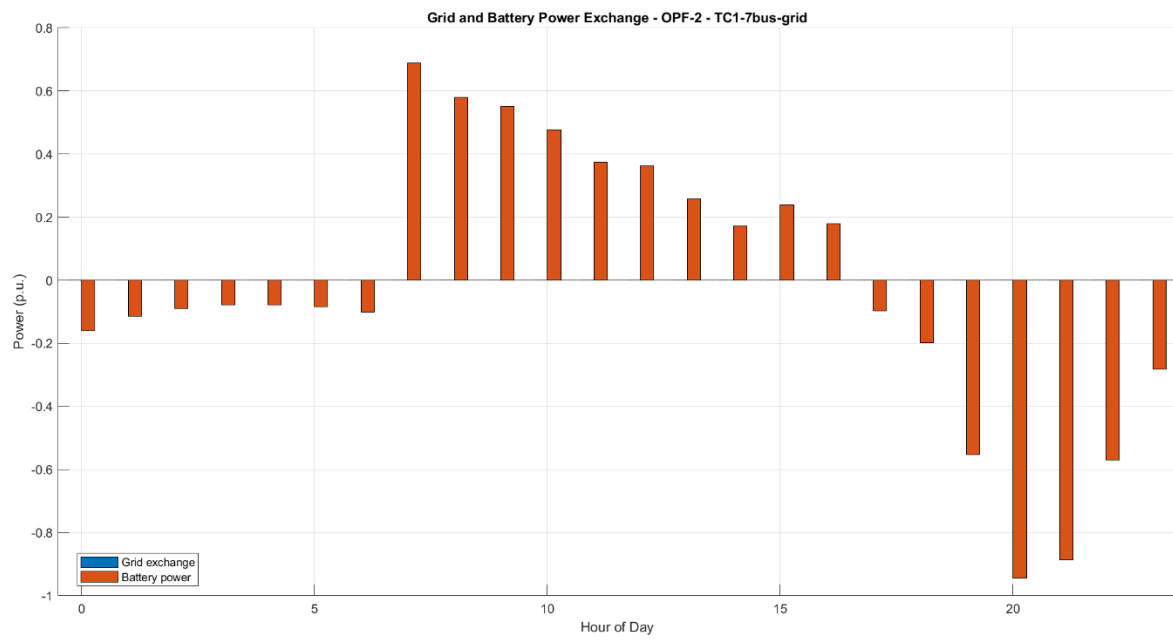
Test ID	Test Type	Description	Input	Expected Result	Actual Result
IT-22	Integration Test	Verify that the user-selected hour_idx is correctly passed into generate_tables	Any run_tcX(...) execution with a valid chosen hour_idx	The selected hour index should be accepted from the command window and used to generate the results table corresponding to that exact hour	The selected hour_idx was successfully accepted and used by generate_tables, producing the results table for the specified hour without errors

Command Window					
Select result hour to generate tables (1 to 24):					
Enter hour index (1-24): 20					
===== TC3-20bus-grid - OPF-2 (Hour 20 snapshot) =====					
Bus No.	Pi (p.u.)	PBi (p.u.)	SOCi (%)	NodalVoltage (p.u.)	Converter Efficiency (%)
1	0.003	-0.119	76.438	1.048	"71.357"
2	0.003	-0.069	75.09	1.047	"73.941"
3	0.004	-0.116	74.596	1.047	"79.787"
4	0.003	-0.061	86.284	1.047	"73.899"
5	0.003	-0.072	75.627	1.048	"71.273"
6	0	-0.079	76.942	1.048	"Non-Op"
7	0.002	-0.077	79.015	1.048	"67.221"
8	0.003	-0.079	74.868	1.048	"71.408"
9	0.002	-0.108	70.649	1.048	"67.150"
10	0	-0.096	70.075	1.048	"Non-Op"
11	0.003	-0.07	73.756	1.048	"72.036"
12	0.003	-0.119	79.051	1.047	"75.266"
13	0.005	-0.068	81.101	1.047	"81.828"
14	0.003	-0.068	78.881	1.047	"74.499"
15	0.003	-0.117	69.05	1.048	"71.815"
16	0.169	-0.248	62.951	1.044	"97.626"
17	0.06	-0.129	82.86	1.046	"96.807"
18	0.058	-0.173	80.259	1.046	"96.758"
19	0.074	-0.183	54.576	1.041	"97.085"
20	-0.399	0	0	1.033	"97.515"

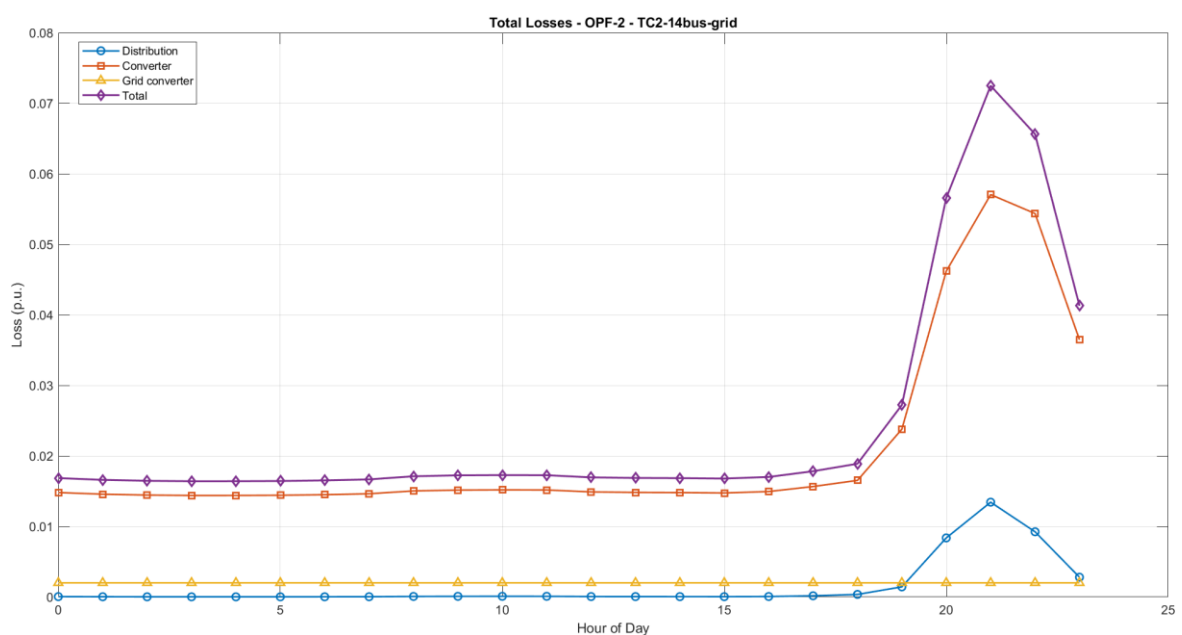
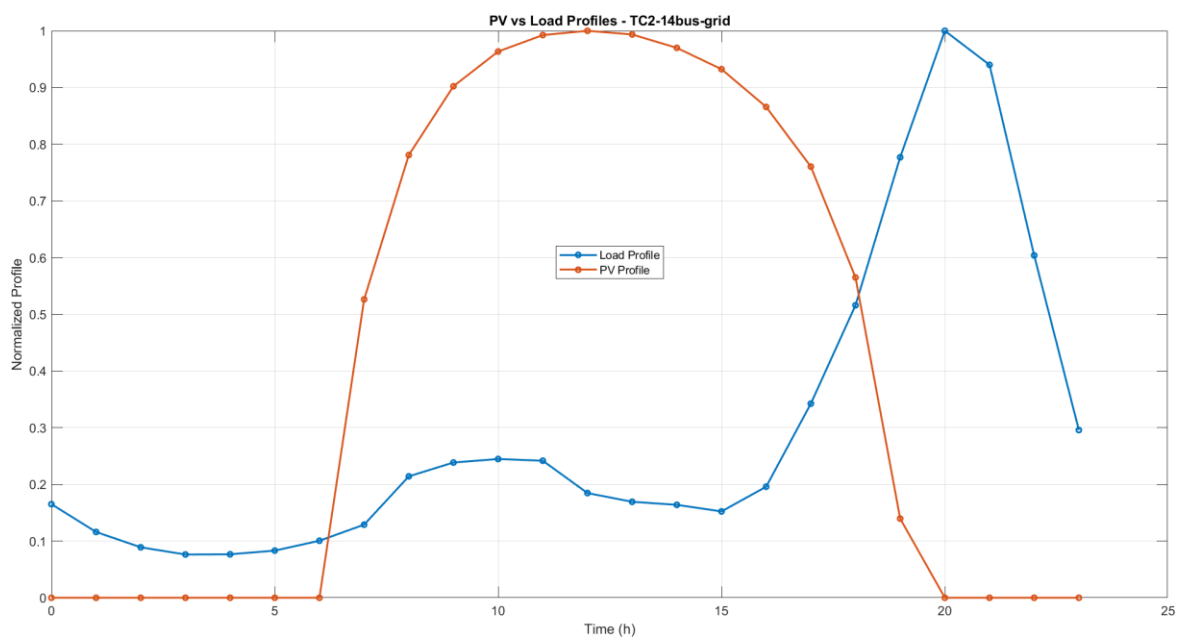
Test ID	Test Type	Description	Input	Expected Result	Actual Result
IT-23	Integration Test	Verify that plot_testcase_figures generates all expected figures for TC1	run_tc1(...)	The plotting routine should generate all expected TC1 visual outputs, including system performance figures such as profiles, losses, dispatch, and grid exchange where applicable	plot_testcase_figures executed successfully for TC1 and generated all expected figures without runtime errors

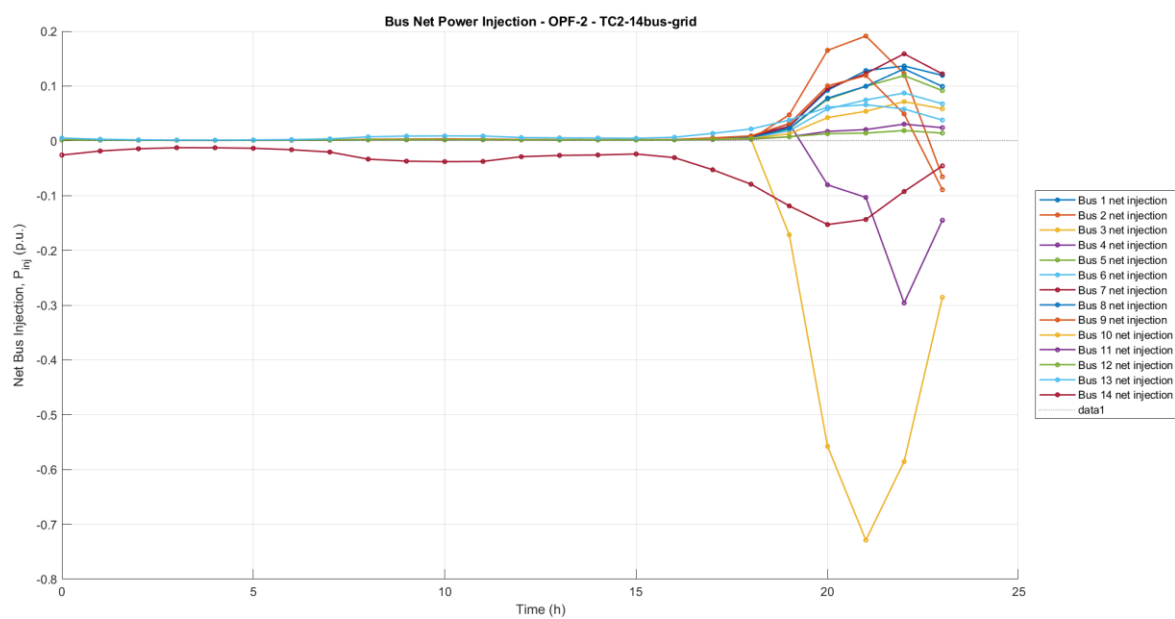
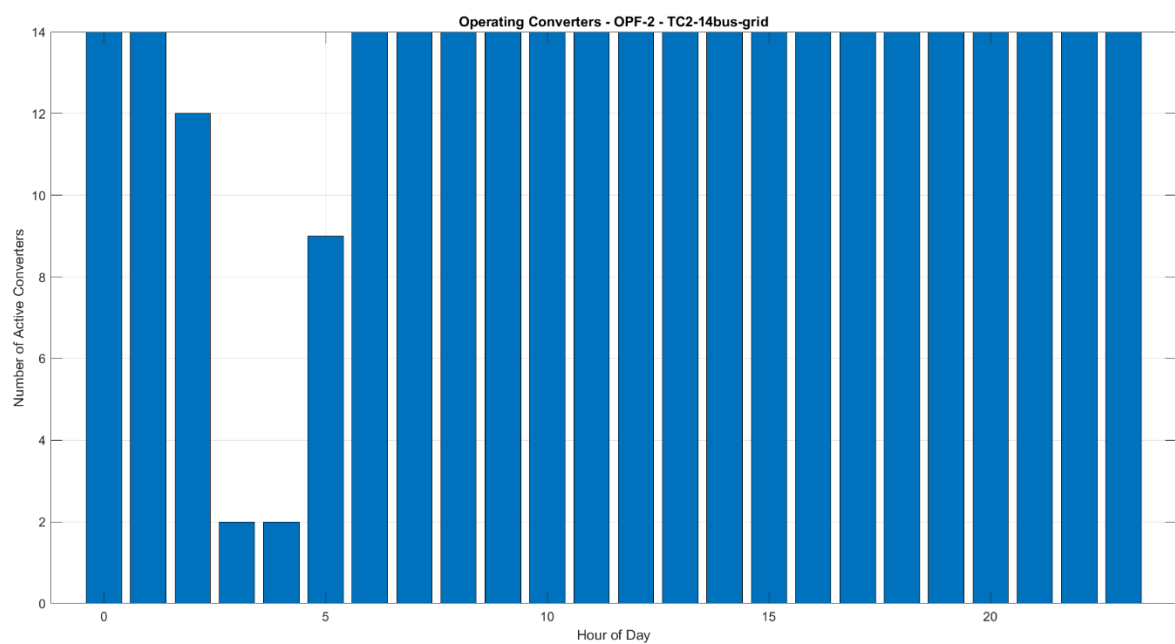


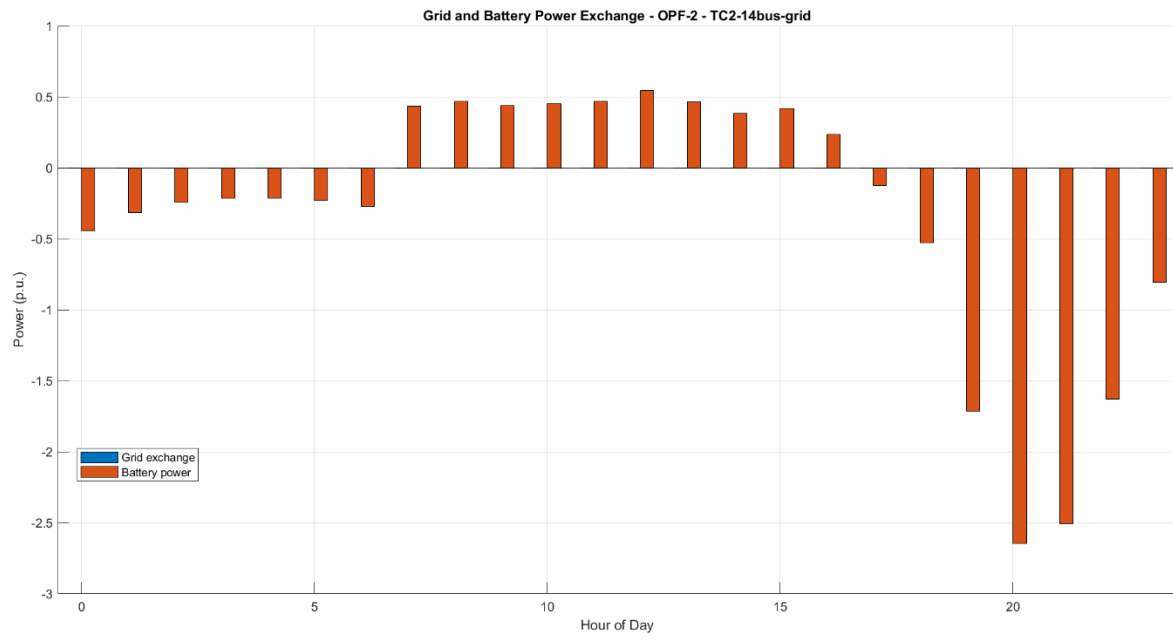




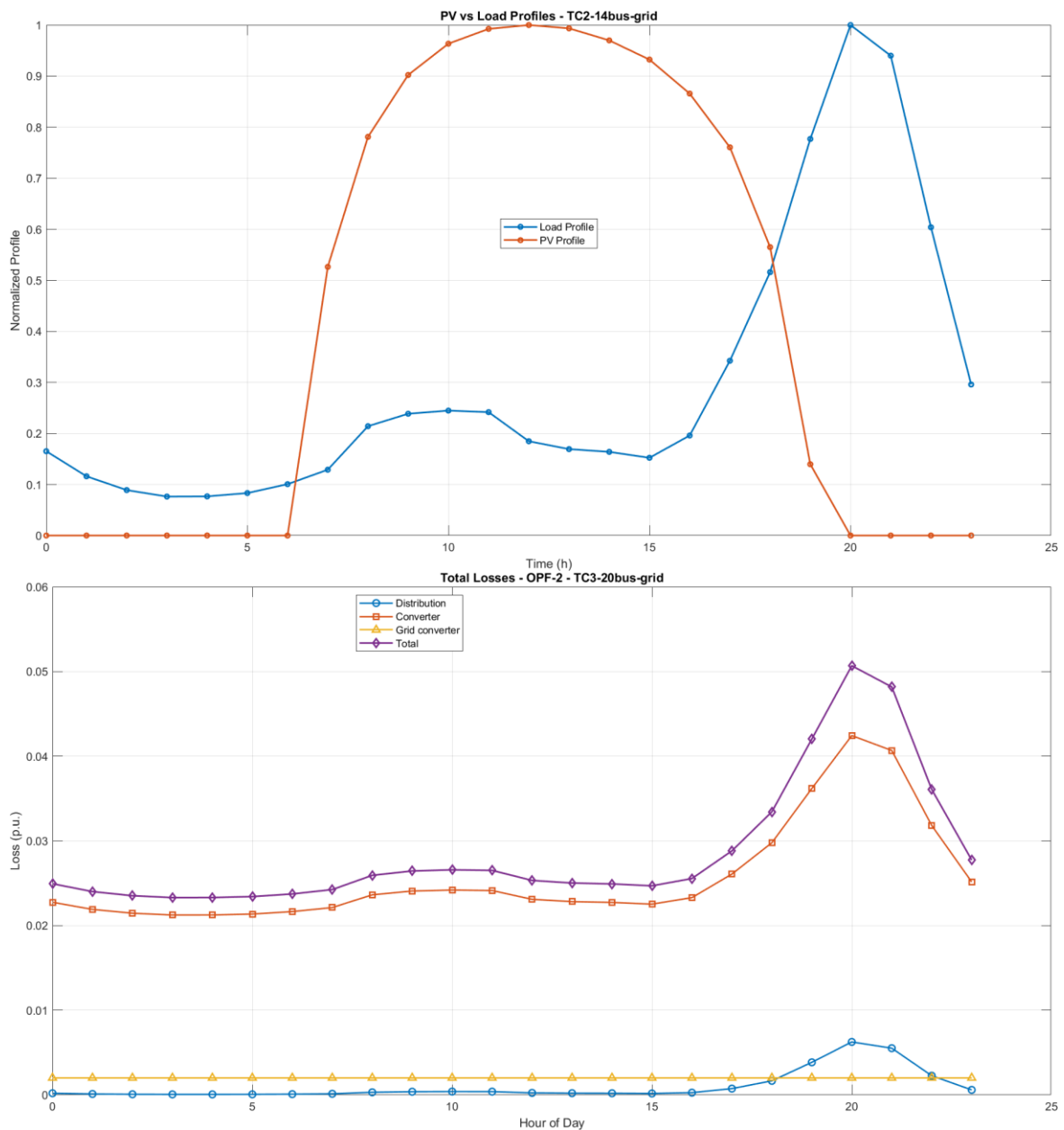
Test ID	Test Type	Description	Input	Expected Result	Actual Result
IT-24	Integration Test	Verify that plot_testcase_figures generates all expected figures for TC2	run_tc2(...)	The plotting routine should generate all expected TC2 visual outputs, including system performance figures such as profiles, losses, dispatch, and grid exchange where applicable	plot_testcase_figures executed successfully for TC2 and generated all expected figures without runtime errors

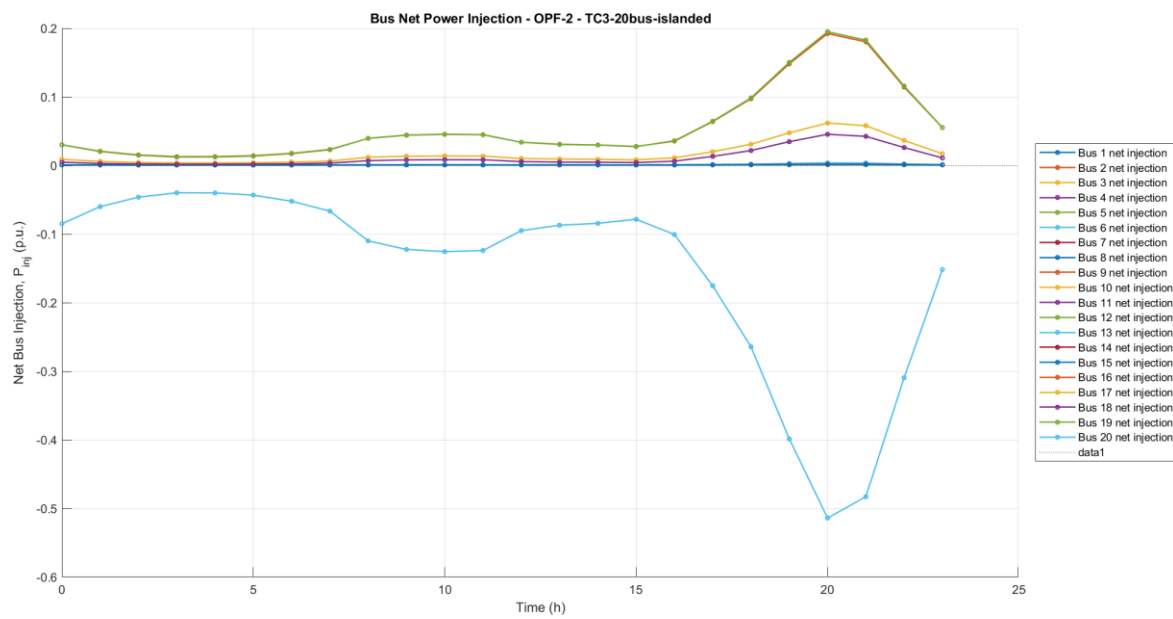


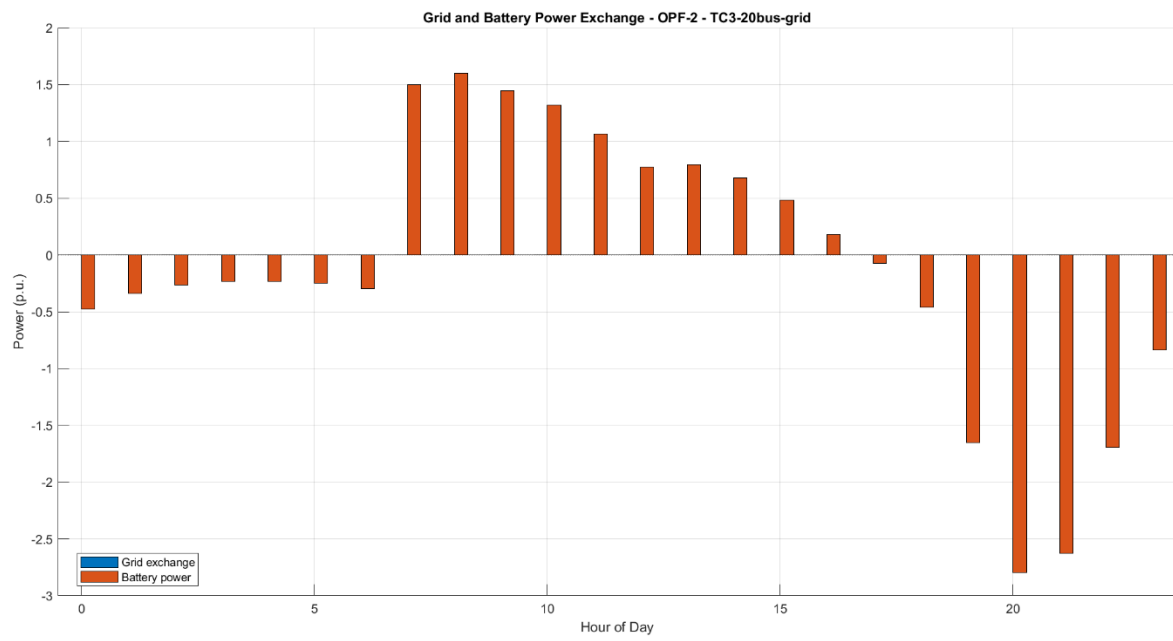




Test ID	Test Type	Description	Input	Expected Result	Actual Result
IT-25	Integration Test	Verify that plot_testcase_figures generates all expected figures for TC3	run_tc3(...)	The plotting routine should generate all expected TC3 visual outputs, including system performance figures such as profiles, losses, dispatch, and grid exchange where applicable	plot_testcase_figures executed successfully for TC3 and generated all expected figures without runtime errors







Test ID	Test Type	Description	Input	Expected Result	Actual Result
VT-01	Validation Test	Verify that an invalid hour_idx below 1 is rejected by the system	Any run_tcX(...) with hour_idx = 0	The system should reject the input and display an error indicating that the hour index must be between 1 and T	The input was rejected successfully. The system returned an error stating that the hour index must be between 1 and 24, preventing invalid table generation

```

===== TC1 OPF2 =====
Grid connected mode: 1
PV availability scale: 1.00
Successful hours: 24/24
Total dist loss (p.u.): 0.018194
Total conv loss (p.u.): 0.244430
Total grid conv loss (p.u.): 0.048014
Total system loss (p.u.): 0.310638
Optimization runtime: 27.2888 seconds

Select result hour to generate tables (1 to 24):
Enter hour index (1-24): 0
Error using run_tc1 (line 61)
Hour index must be between 1 and 24.

```

Test ID	Test Type	Description	Input	Expected Result	Actual Result
VT-02	Validation Test	Verify that an invalid hour_idx greater than T is rejected by the system	Any run_tcX(...) with hour_idx = 25	The system should reject the input and display an error indicating that the hour index must be between 1 and T	The input was rejected successfully. The system returned an error stating that the hour index must be between 1 and 24, preventing invalid table generation

```

===== TC1 OPF2 =====
Grid connected mode: 1
PV availability scale: 1.00
Successful hours: 24/24
Total dist loss (p.u.): 0.018194
Total conv loss (p.u.): 0.244430
Total grid conv loss (p.u.): 0.048014
Total system loss (p.u.): 0.310638
Optimization runtime: 32.1261 seconds

Select result hour to generate tables (1 to 24):
Enter hour index (1-24): 25
Error using run_tc1 (line 61)
Hour index must be between 1 and 24.

fx >>

```

Test ID	Test Type	Description	Input	Expected Result	Actual Result
VT-03	Validation Test	Verify that a nonnumeric or invalid hour_idx input is not accepted	Any run_tcX(...) with nonnumeric hour_idx input	The system should reject the input and return an error because the hour index must be a single finite numeric value	The invalid input was rejected successfully. The system returned an error indicating that there was not enough input arguments

```

===== TC1 OPF2 =====
Grid connected mode: 1
PV availability scale: 1.00
Successful hours: 24/24
Total dist loss (p.u.): 0.018194
Total conv loss (p.u.): 0.244430
Total grid conv loss (p.u.): 0.048014
Total system loss (p.u.): 0.310638
Optimization runtime: 24.8013 seconds

Select result hour to generate tables (1 to 24):
Enter hour index (1-24): cat
Error using cat
Not enough input arguments.

```

Test ID	Test Type	Description	Input	Expected Result	Actual Result
VT-04	Validation Test	Verify that PV availability values below 0 are clamped to the minimum valid bound	Any run_tcX(...) with pv_scale = -0.5	The system should automatically clamp the PV availability to 0.0 and continue execution without failure	The PV availability input was clamped successfully to 0.0. The simulation continued normally, and the system operated with zero PV contribution as intended

```

Set PV availability level for TC1 (0 to 1):
  1 = Full PV generation
  0 = No PV generation
Enter PV availability (0-1): -0.5
Reading load data from sheet: Test Case 1
Successfully read load data from sheet: Test Case 1

===== TC1 OPF2 =====
Grid connected mode: 1
PV availability scale: 0.00
Successful hours: 24/24
Total dist loss (p.u.): 0.027402
Total conv loss (p.u.): 0.260920
Total grid conv loss (p.u.): 0.069183
Total system loss (p.u.): 0.357504
Optimization runtime: 21.8204 seconds

```

Test ID	Test Type	Description	Input	Expected Result	Actual Result
VT-05	Validation Test	Verify that PV availability values above 1 are clamped to the maximum valid bound	Any run_tcX(...) with pv_scale = 1.5	The system should automatically clamp the PV availability to 1.0 and continue execution without failure	The PV availability input was clamped successfully to 1.0. The simulation continued normally, and the full PV profile was applied as intended

```

Set PV availability level for TC1 (0 to 1):
  1 = Full PV generation
  0 = No PV generation
Enter PV availability (0-1): 1.5
Reading load data from sheet: Test Case 1
Successfully read load data from sheet: Test Case 1

===== TC1 OPF2 =====
Grid connected mode: 1
PV availability scale: 1.00
Successful hours: 24/24
Total dist loss (p.u.): 0.018194
Total conv loss (p.u.): 0.244430
Total grid conv loss (p.u.): 0.048014
Total system loss (p.u.): 0.310638
Optimization runtime: 25.3796 seconds

```